

QIQY5

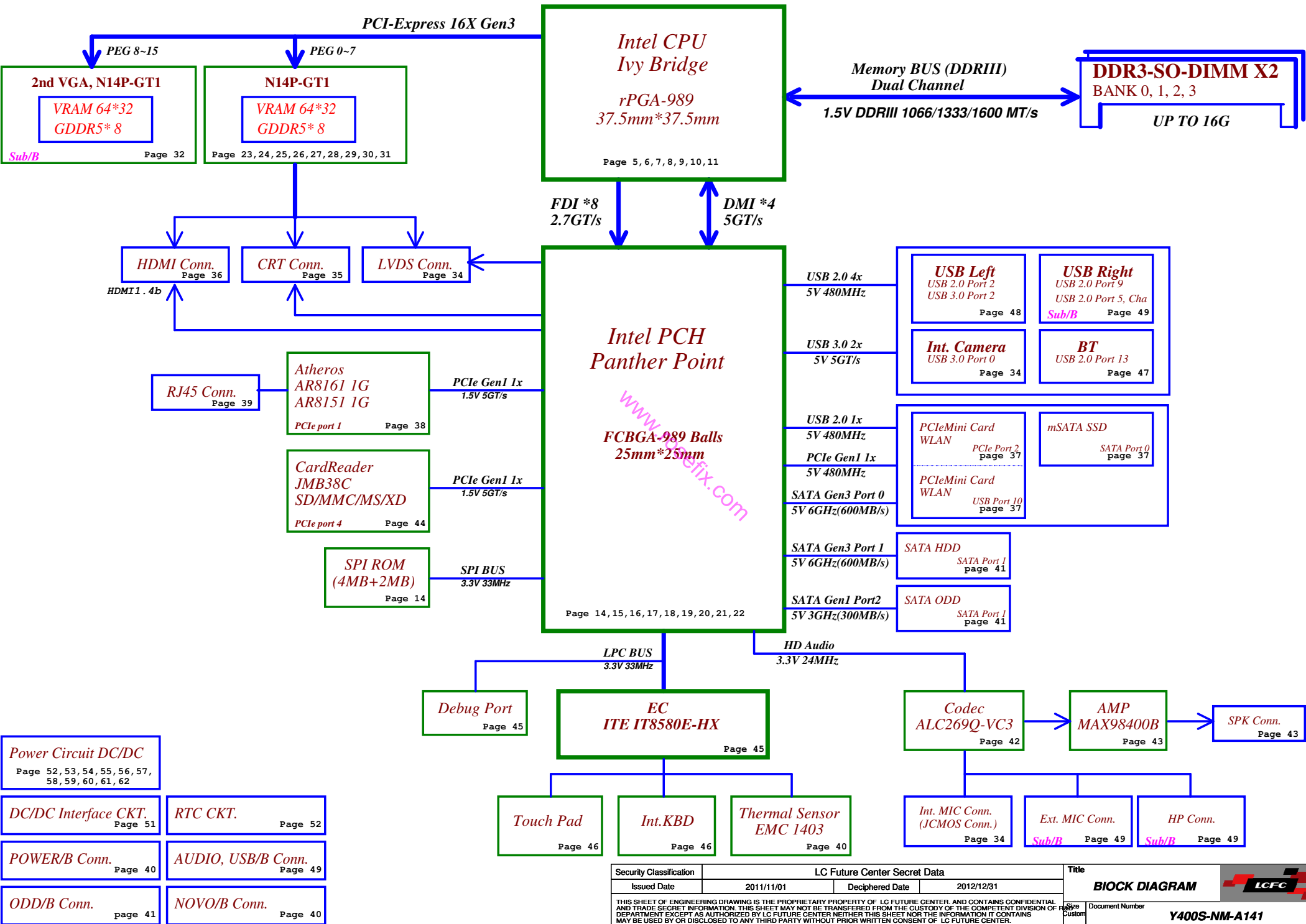
Whisky3.0 (Y400S)

NM-A141 Rev0.2 Schematic

**Intel IVY Bridge Processor with DDRIII + Panther Point PCH
nVIDIA N14P GT + 2nd VGA N14P GT**

2012-10-25-Rev0.2

Security Classification	LC Future Center Secret Data			Title	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	COVER PAGE	
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Voltage Rails (O --> Means ON , X --> Means OFF)

Power Plane / State	B+	+3VALW +5VALW	+1.5V	+5VS +3VS +1.5VS +VCCSA +V1.5S_VCCP +CPU_CORE +VGA_CORE +GFX_CORE +1.8VS +1.05VS +0.75VS +3.3VS_VGA +1.5VS_VGA +1.05VS_VGA
S0	O	O	O	O
S3	O	O	O	X
S5 S4/AC Only	O	O	X	X
S5 S4 Battery only	O	X	X	X
S5 S4 AC & Battery don't exist	X	X	X	X

STATE \ SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1 (Power On Suspend)	LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)	LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)	LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)	LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

USB Port Table

USB 2.0	USB 3.0	Port	4 External USB Port
			Camera
	XHCI 1	0	
		1	
		2	USB Port (Left Side)
		3	
		4	
		5	USB Port (Right Side)
		6	
		7	
		8	
		9	USB Port (Right Side)
		10	Mini Card(WLAN)
		11	
		12	
		13	Blue Tooth

BOM Structure Table

BOM Structure	BTO Item
HDMI@	HDMI part
CHG@	USB charger part
NOCHG@	No USB charger part
CMOS@	CMOS Camera part
8161@	AR8161 LAN part
8151@	AR8151 LAN part
8161S@	AR8161 LAN surge part
8151S@	AR8151 LAN surge part
SURGE@	AR8151/8161 LAN surge part --> Delete (201200627)
61@	X76 P/N for AR8161
51@	X76 P/N for AR8151
X76@	X76 Level part for VRAM
GC6@	NV CG6 support part
NOGC6@	NV no CG6 support part
AOAC@	AOAC support part
KBL@	K/B Light part
ME@	ME part
OPT@	For optimus function part
SLI@	For SLI function part
DS3@	Deep S3 support part
S3@	For S3 function part
GT@	NV chip part
@	Unpop

SMBUS Control Table

	SOURCE	Main VGA	2nd VGA	BATT	IT8580E	SODIMM	WLAN WiMAX	Thermal Sensor	PCH	TP Module
EC_SMB_CK1 EC_SMB_DA1	IT8580E +3VALW	X	X	V +3VALW	X	X	X	X	X	X
EC_SMB_CK2 EC_SMB_DA2	IT8580E +3VS	V +3VS	V +3VS	X	X	X	X	V +3VS	V +3V_PCH	X
SMB_CLK_S3 SMB_DATA_S3	PCH +3VS	X	X	X	X	V +3VS	V +3VS	X	V +3V_PCH	V +3VS

PCIe PORT LIST

Port	Device
1	LAN
2	WLAN
3	
4	Card Reader
5	
6	
7	
8	

EC SM Bus1 address

Device	Address
Smart Battery	0001 011X b

EC SM Bus2 address

Device	Address
Thermal Sensor EMC1403-2	1001_101xb
Master VGA	0x9E
Slave VGA	0x9C

PCH SM Bus address

Device	Address
DDR DIMM0	1001 000Xb
DDR DIMM2	1001 010Xb



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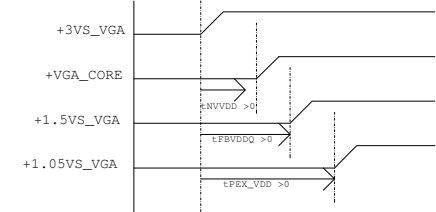
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NOTES LIST	Y400S-NM-A141	1.0
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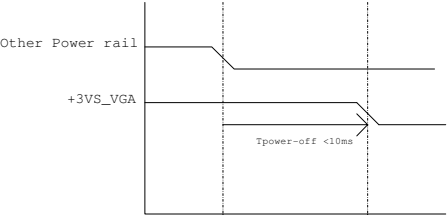
Hot plug detect for IFP link E

VGA and GDDR5 Voltage Rails (N13Px GPIO)

GPIO	I/O	ACTIVE	Function Description
GPIO0	OUT	-	FB_CLAMP
GPIO1	OUT	-	
GPIO2	OUT	-	VGA_BL_PWM
GPIO3	OUT	-	VGA_ENVDD
GPIO4	OUT	-	VGA_ENBKL
GPIO5	OUT	-	
GPIO6	OUT	-	FB_CLAMP_TOGGLE_REQ#
GPIO7	OUT	-	
GPIO8	I/O	-	OVERT#
GPIO9	OUT	-	VGA_ALERT#
GPIO10	OUT	-	Memory VREF Control
GPIO11	OUT	-	NVVD PWM_VID
GPIO12	IN		VGA_AC_DET_R (10K pull High)
GPIO13	OUT	-	DPRS LPVR_VGA
GPIO14	OUT	-	
GPIO15	IN	N/A	
GPIO16	OUT	-	
GPIO17	IN	N/A	
GPIO18	IN	-	dGPU_HDMI_HPD
GPIO19	IN	-	



1. all power rail ramp up time should be larger than 40us



1. all GPU power rails should be turned off within 10ms
2. Optimus system VDD33 avoids drop down earlier than NVDD and FBVDDQ

Performance Mode P0 TDP at Tj = 102 C* (GDDR5)

	GPU (4)	Mem (1,5)	NVCLK /MCLK	NVVDD			FBVDD (1.35V)		FBVDDQ (GPU+Mem) (1.35V)		PCI Express (1.05V)		I/O and PLLVDD (1.8V)		I/O and PLLVDD (1.05V)		Other (3.3V)	
Products	(W)	(W)	(MHz)	(V)	(A)	(W)	(A)	(W)	(A)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)
N13X 128bit 1GB GDDR5	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

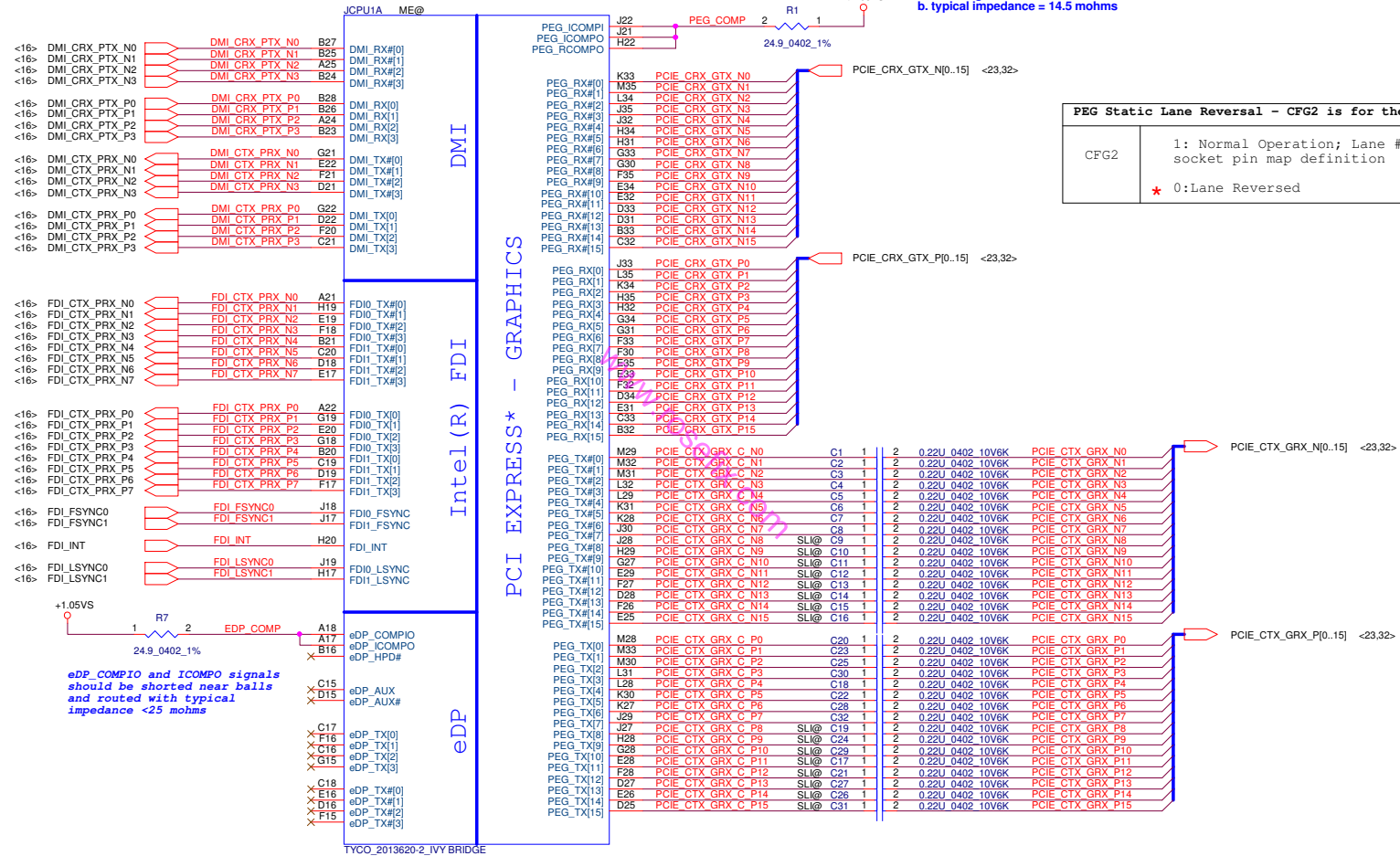
Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

	Device ID		setting	I2C Slave addresses ID
N13P-GT (28nm)	0x0FDB	SMB_ALT_ADDR (ROM_SO Bit 1)	0	0x9E
			1	0x9C

GPU	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4	
N13P-GT1 28nm	PU 10K	PU 25K	PU 45K	PD 35K	PD 10K	PU 5K	PD 10K	Master
	PU 20K	PU 25K	PU 45K	PD 35K	PD 10K	PD 5K	PD 10K	Slave

GPU		N13P-GT		
FB Memory (GDDR5)		ROM_SI		
Samsung 2500MHz	K4G10325FG-HC04			
	32Mx32	PD 45K		
Hynix 2500MHz	H5GQ1H24BFR-T2C			
	32Mx32	PD 35K		
Samsung 2500MHz	K4G20325FD-FC04			
	64Mx32	PD 30K		
Hynix 2500MHz	H5GQ2H24MFR-T2C			
	64Mx32	PD 25K		

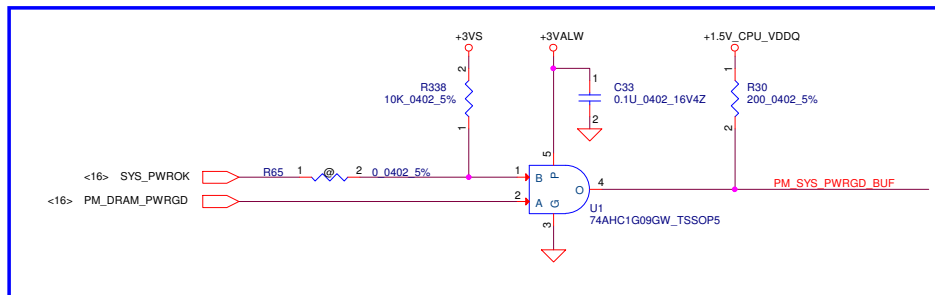
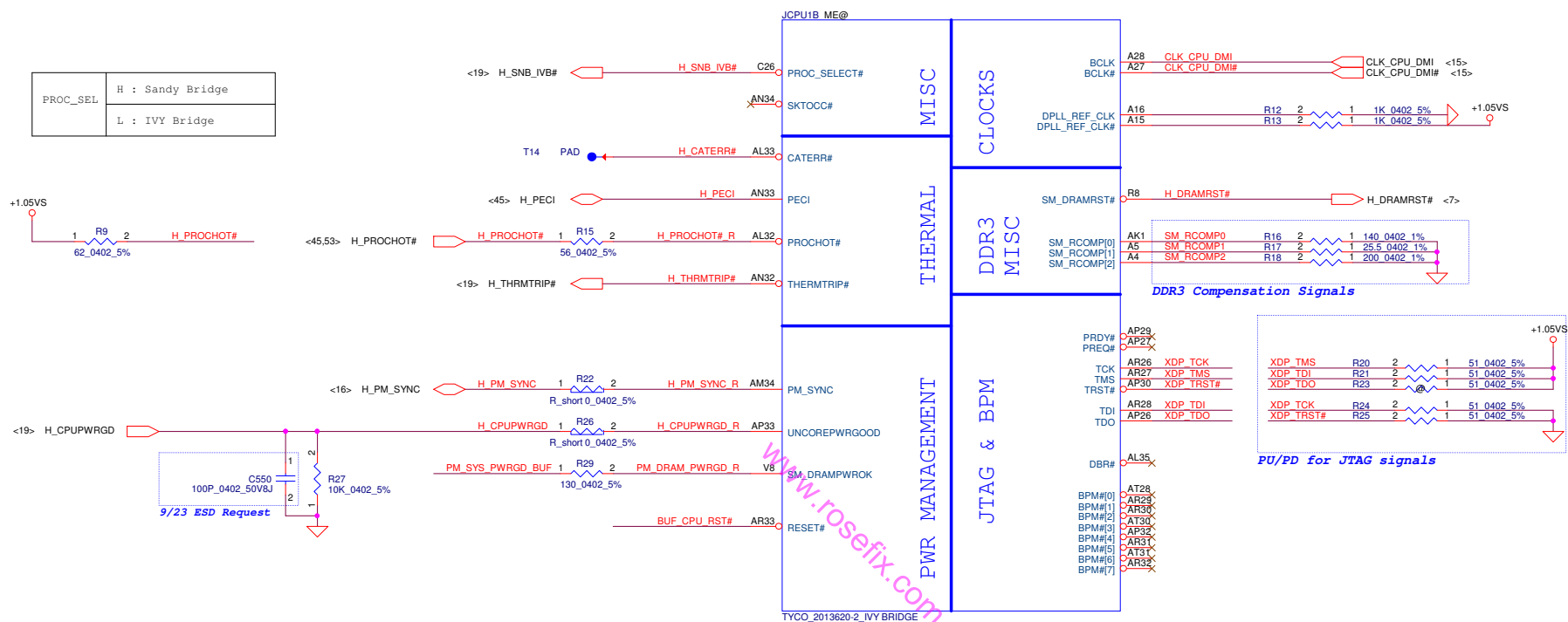
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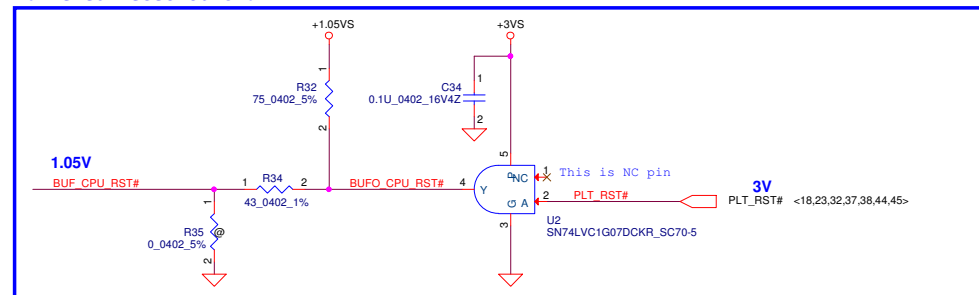
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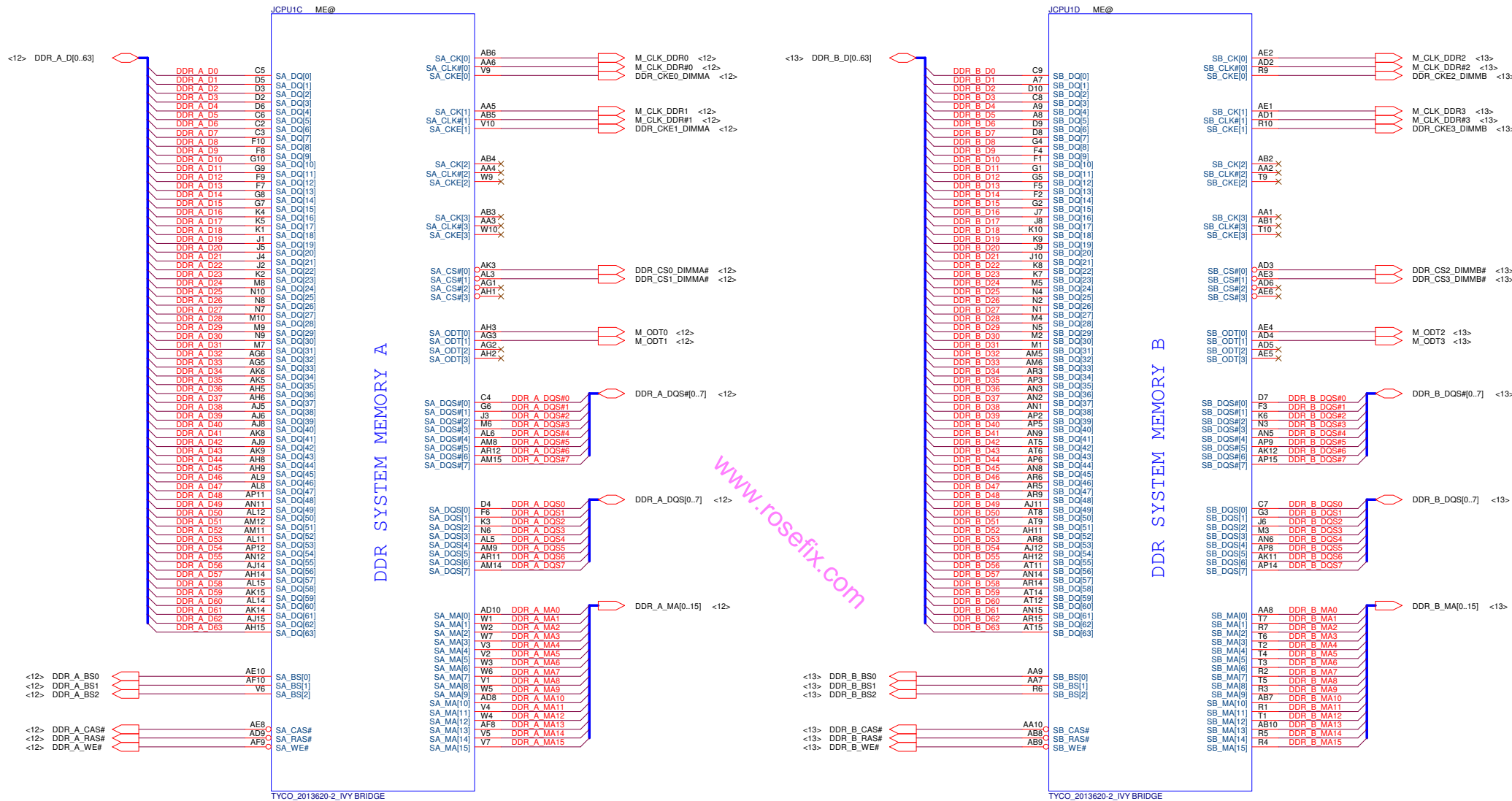
Title	Document Number	Rev
PROCESSOR(1/7) DMI, FDI, PE	Y400S-NM-A141	1.0
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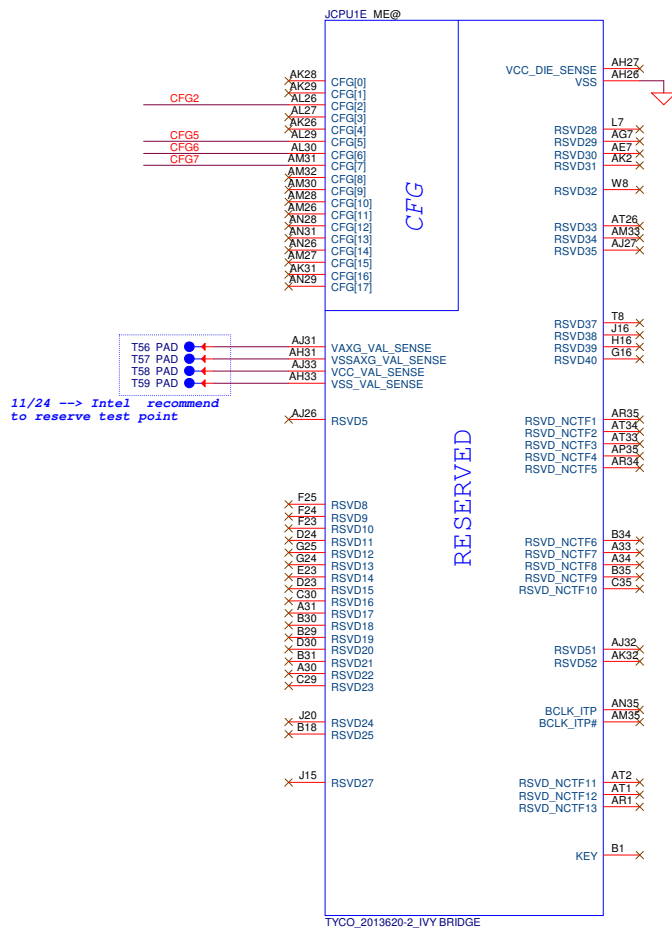
PROC_SEL	H : Sandy Bridge
	L : IVY Bridge



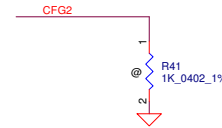
Buffered Reset to CPU







CFG Straps for Processor

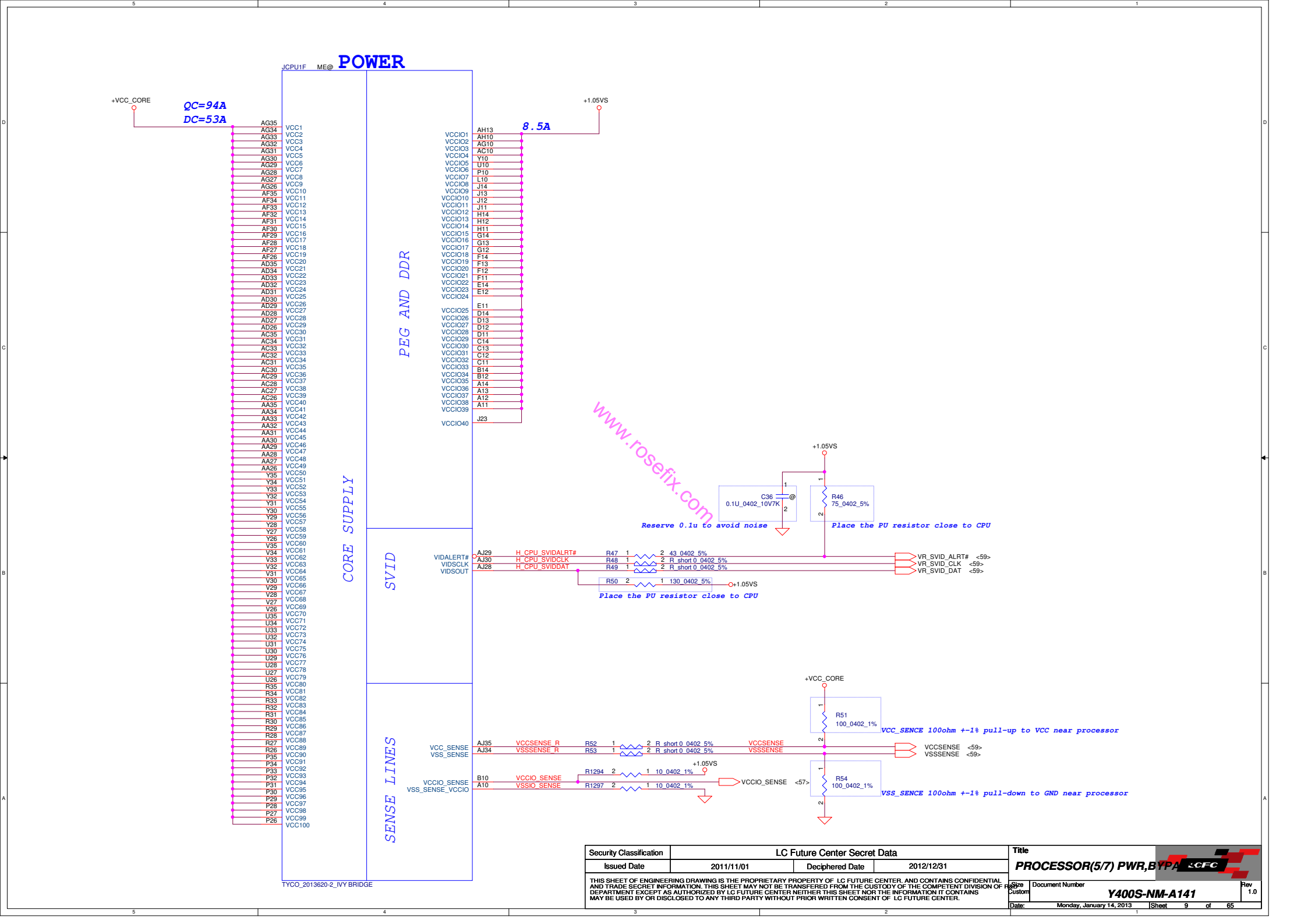


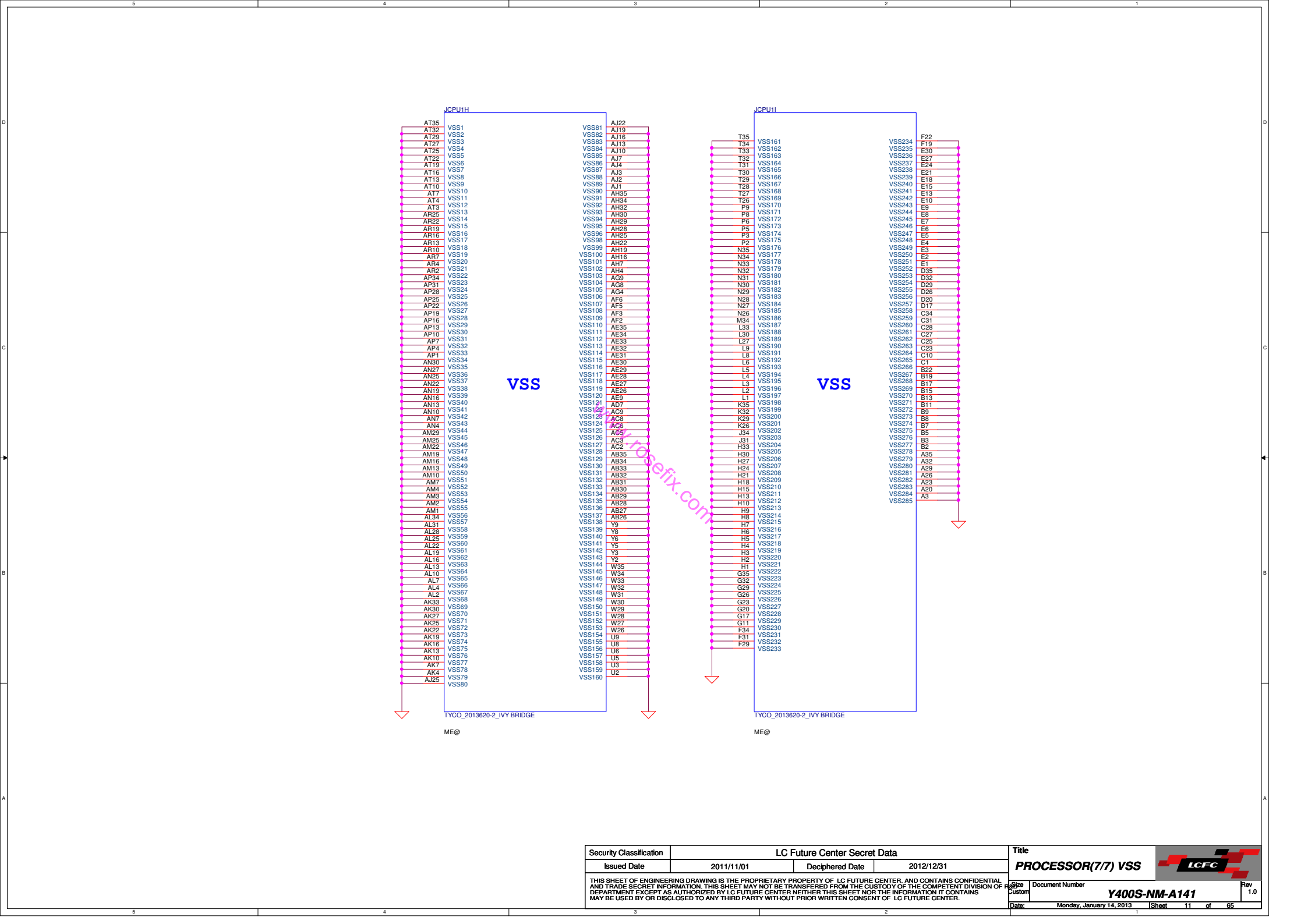
PEG Static Lane Reversal - CFG2 is for the 16x	
CFG2	★ 1: Normal Operation; Lane # definition matches socket pin map definition 0: Lane Reversed

Display Port Presence Strap	
CFG4	★ 1: Disabled; No Physical Display Port attached to Embedded Display Port 0: Enabled; An external Display Port device is connected to the Embedded Display Port

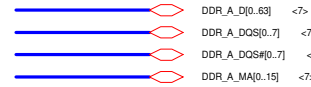
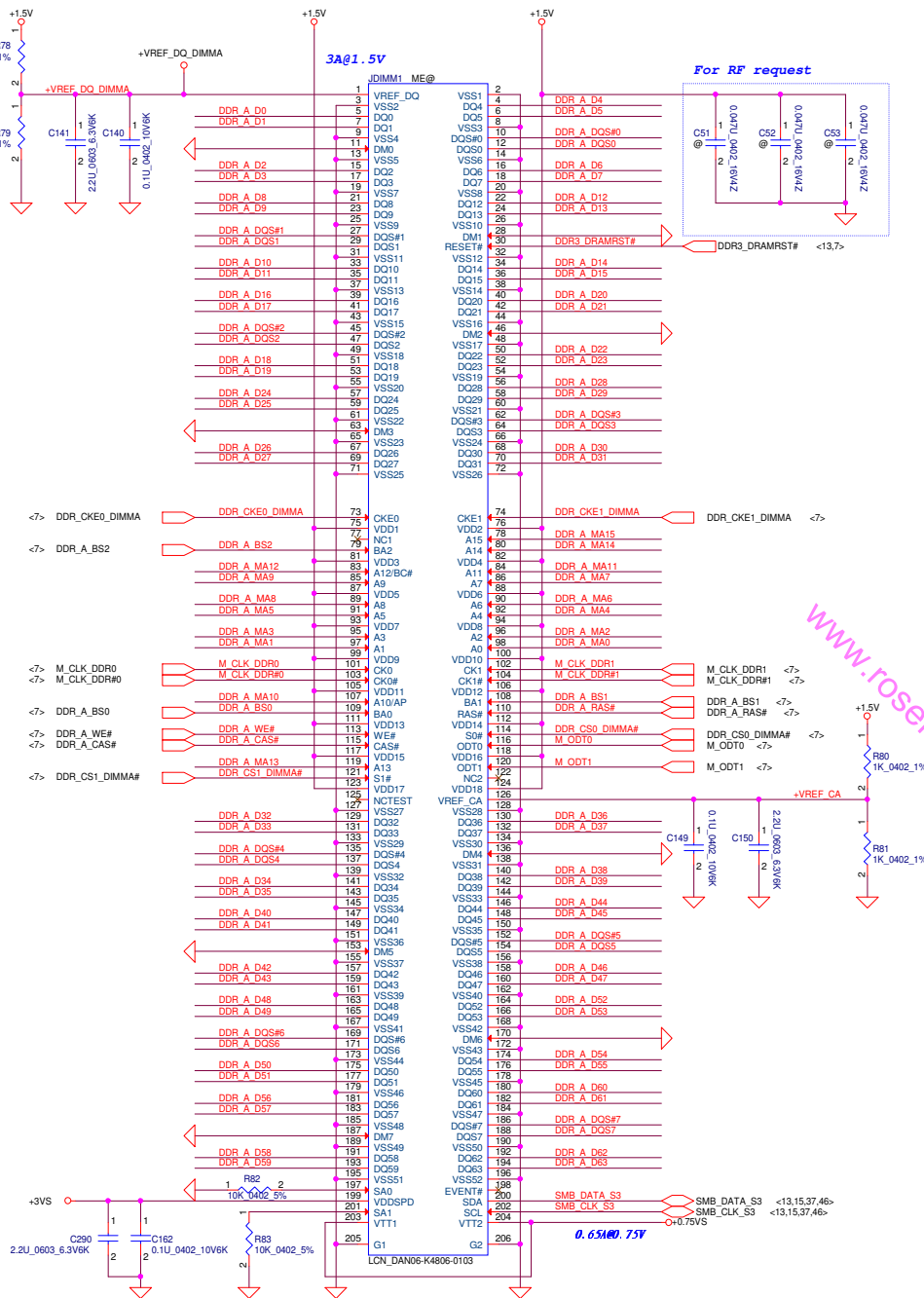
PCIe Port Bifurcation Straps	
CFG[6:5]	11: (Default) x16 - Device 1 functions 1 and 2 disabled ★ 10: x8, x8 - Device 1 function 1 enabled ; function 2 disabled 01: Reserved - (Device 1 function 1 disabled ; function 2 enabled) 00: x8, x4, x4 - Device 1 functions 1 and 2 enabled

PEG DEFER TRAINING	
CFG7	1: (Default) PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training





DDR3 SO-DIMM A



DDR_A_D[0..63] <7>

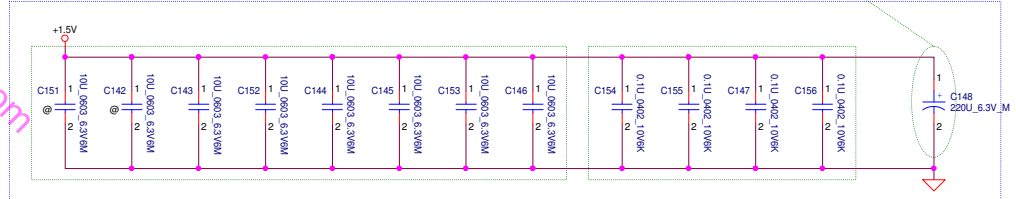
DDR_A_DQS[0..7] <7>

DDR_A_DQS[0..7] <7>

DDR_A_MA[0..15] <7>

Layout Note:
Place near DIMM

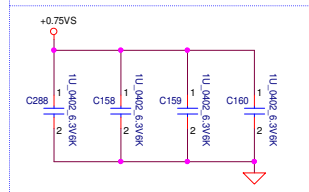
OSCON (220uF 6.3V 4.2L ESR17m) *1=(SF000002Y00)
(10uF_0603_6.3V)*8
(0.1uF_402_10V)*4



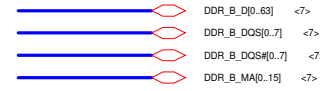
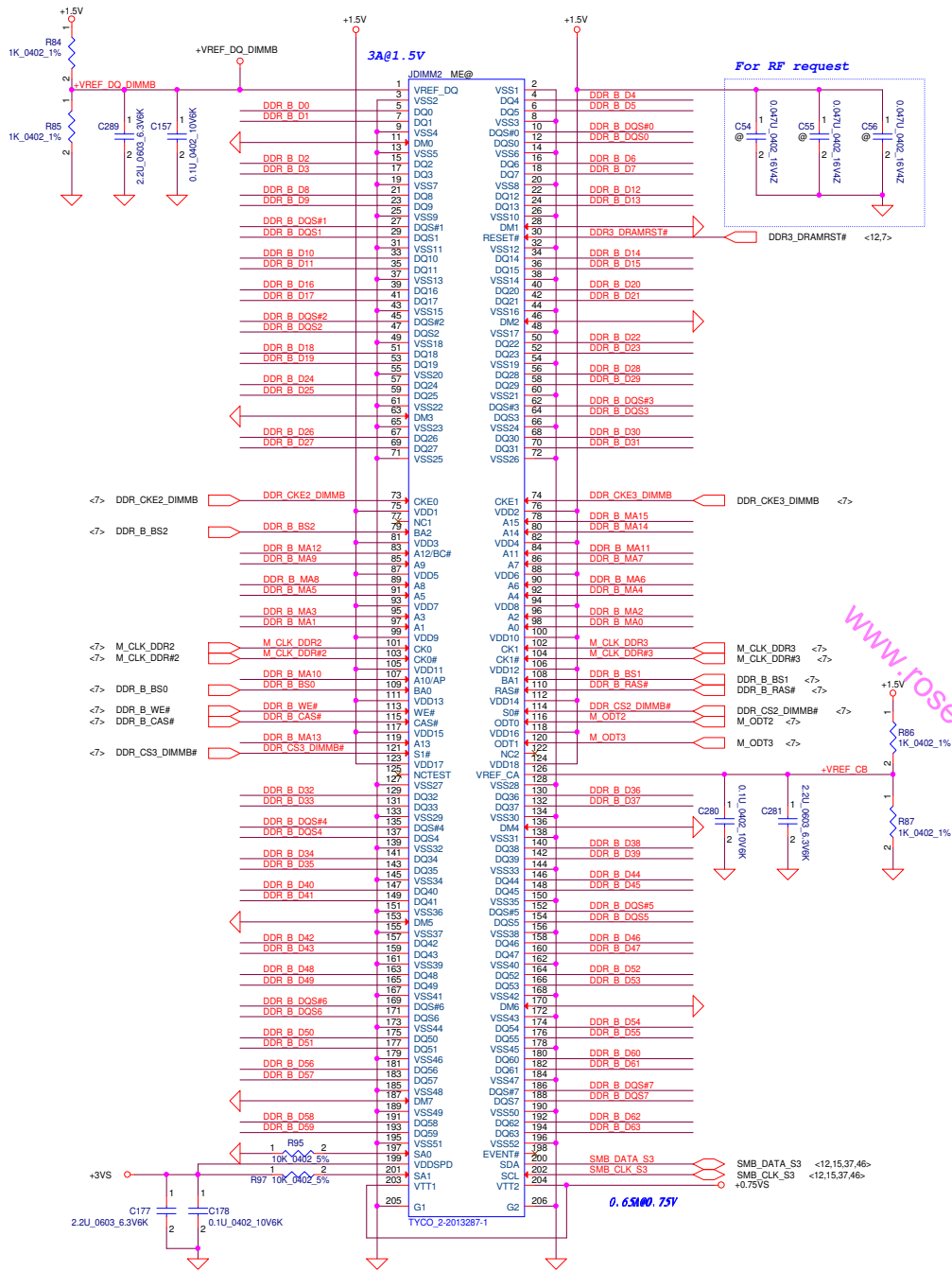
Layout Note:
Place near DIMM

Layout Note:
Place near DIMM

DDR_A_DM[0..7] connect to GND



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DDR3 SO-DIMM B

DDR_B_D[0..63] <7>

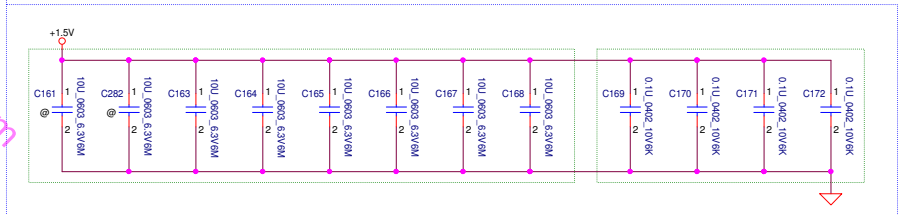
DDR_B_DQS[0..7] <7>

DDR_B_DQS#[0..7] <7

DDR_B_MA[0..15] <7>

Layout Note:
Place near DIMM

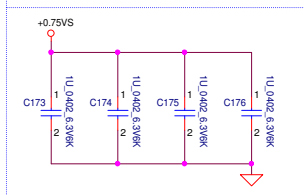
$(10\mu F_{0603_{6.3V}}) * 8$

$$(0.1\mu F_{-402_{-10V}}) * 4$$


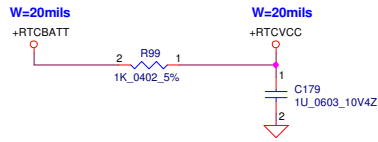
Layout Note:
Place near DIMM

Layout Note:
Place near DIMM

DDR_B_DM[0:7] connect to GND

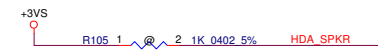


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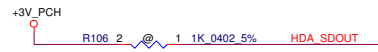


INTVRMEN

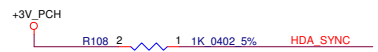
★ H : Integrated VRM enable (Default)
L : Integrated VRM disable
(INTVRMEN should always be pull high.)



HIGH= Enable (No Reboot)
★ LOW= Disable (Default)

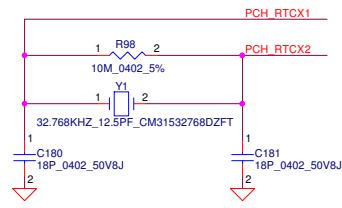
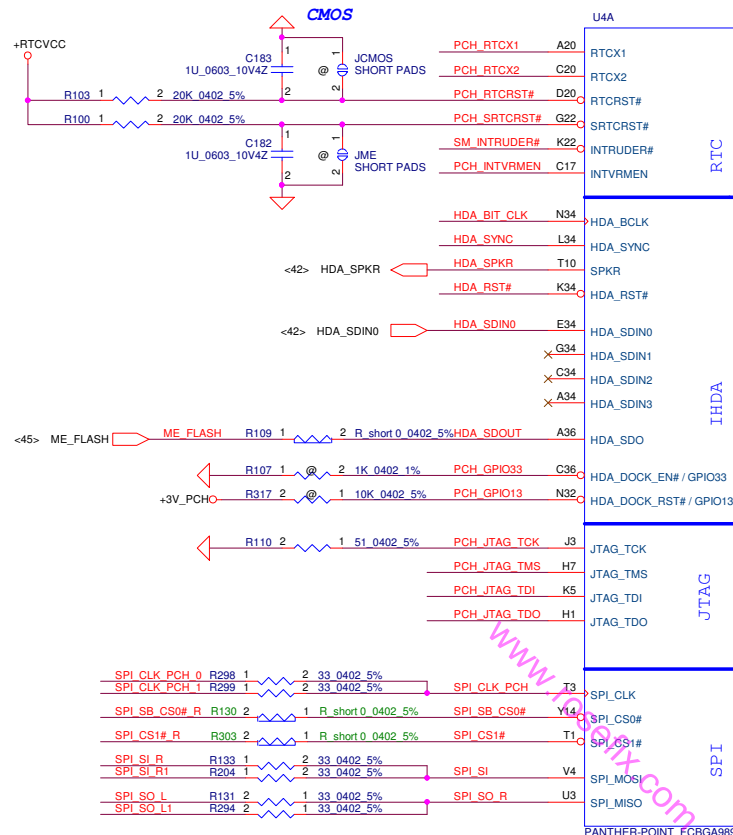
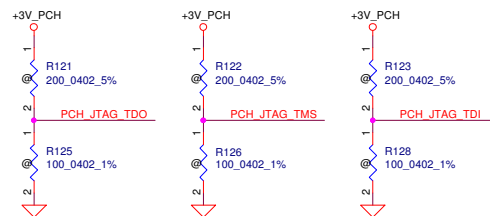
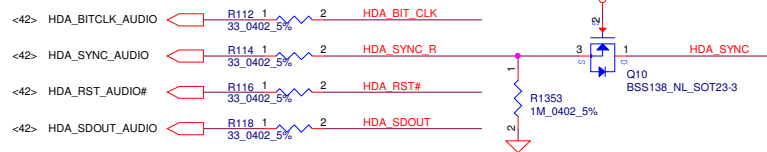


★ Low = Disabled (Default)
High = Enabled [Flash Descriptor Security Override]



This signal has a weak internal pull-down
On Die PLL VR Select is supplied by 1.5V when snapled high (Default)
★ 1.8V when sampled low
Needs to be pulled High for Chief River platform

HDA AUDIO

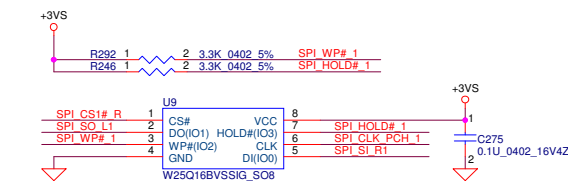
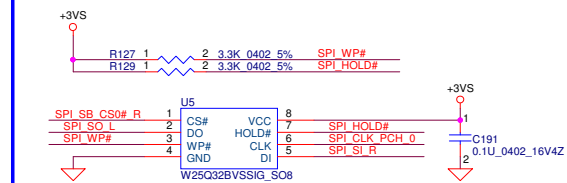


For EMI

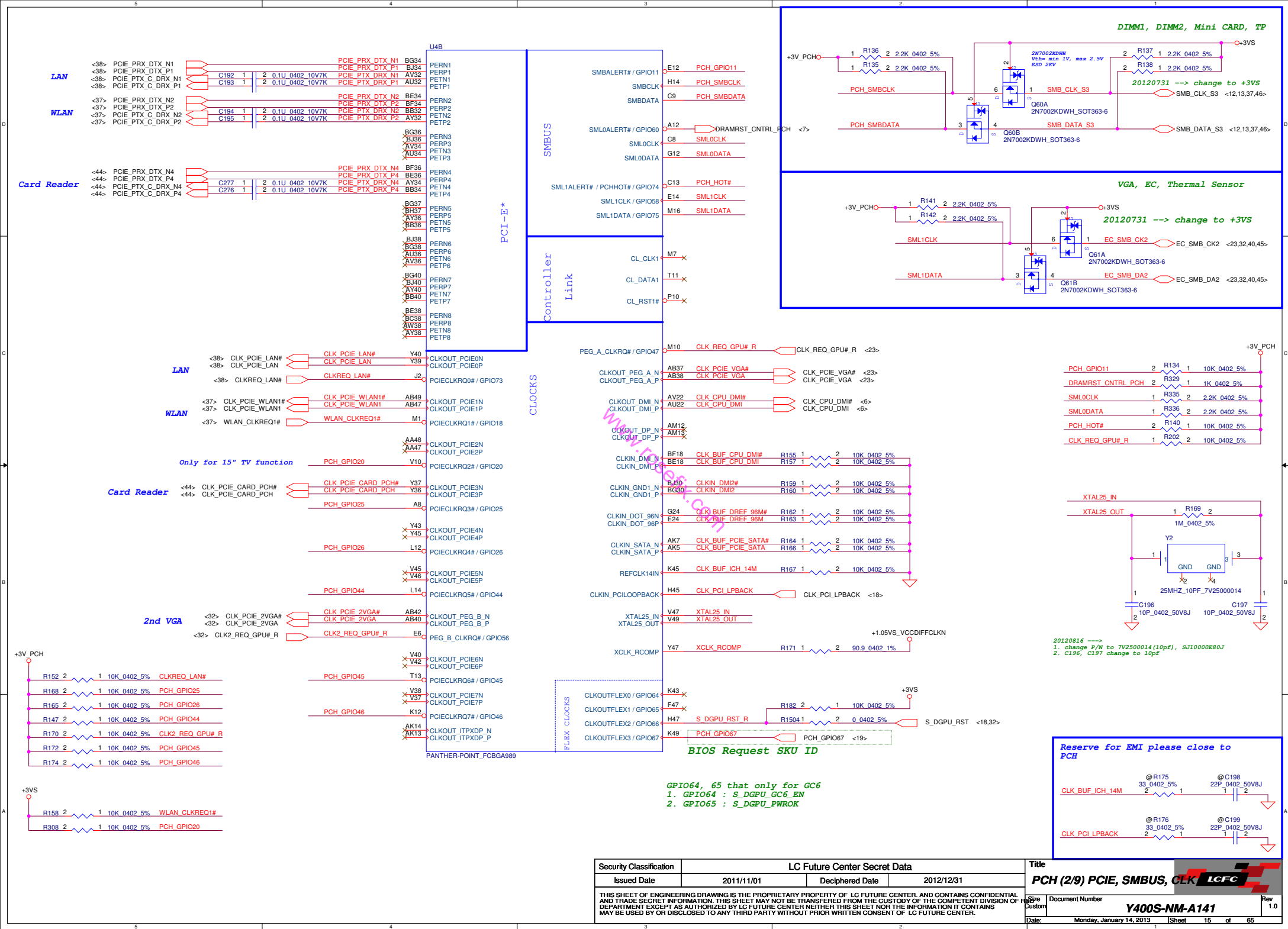


4MB P/N : SA0005P500

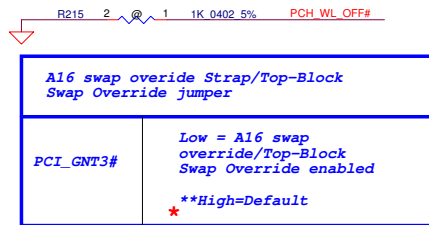
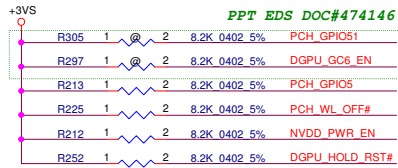
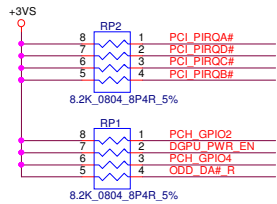
2MB P/N : SA0003FO10



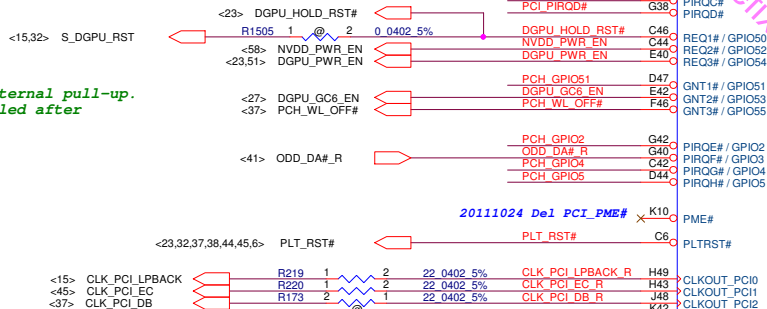
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GPI053 => This Signal has a weak internal pull-up.
NOTE: The internal pull-up is disabled after PLTRST# deasserts.

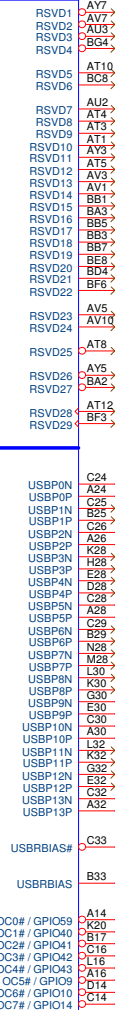
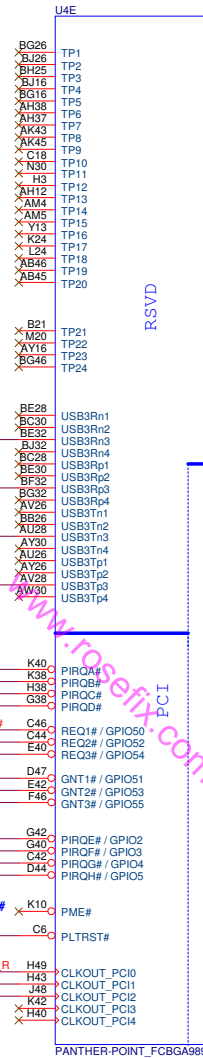


The diagram illustrates the hardware connection for the Boot BIOS Strap. A red line represents the signal path starting from PCH GPIO51, passing through a resistor labeled R221, and then through a 1K resistor to the 0402 pin of the BBS1 component. A red arrow points to the BBS1 component.

Boot BIOS Strap bit1 BBS1		
	Bit11	Bit10
GNT1#/ GPIO51	0	1
	1	0
	1	1
	0	0

		Boot BIOS Destination
	Reserved	
	Reserved	
	* SPI	(Default)
	LPC	

USB3.0	
Port1	
Port2	
Port3	LEFT USB
Port4	



USB DEBUG = PORT1 AND PORT9

Camera

LEFT USB

RIGHT USB 1 (CHARGER PORT, SUB/B)

Some PCH config not support USB port 6 & 7.

RIGHT USB 2 (SUB/B)

WLAN

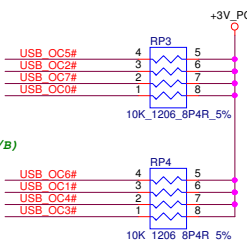
BT

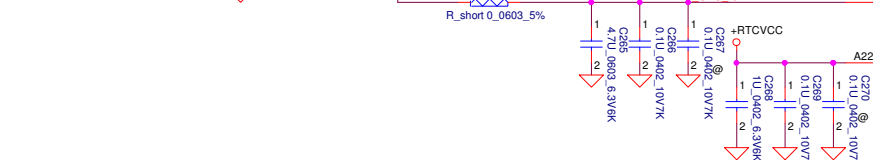
Within 500 mils

USB3 Port3, USB2 Port2

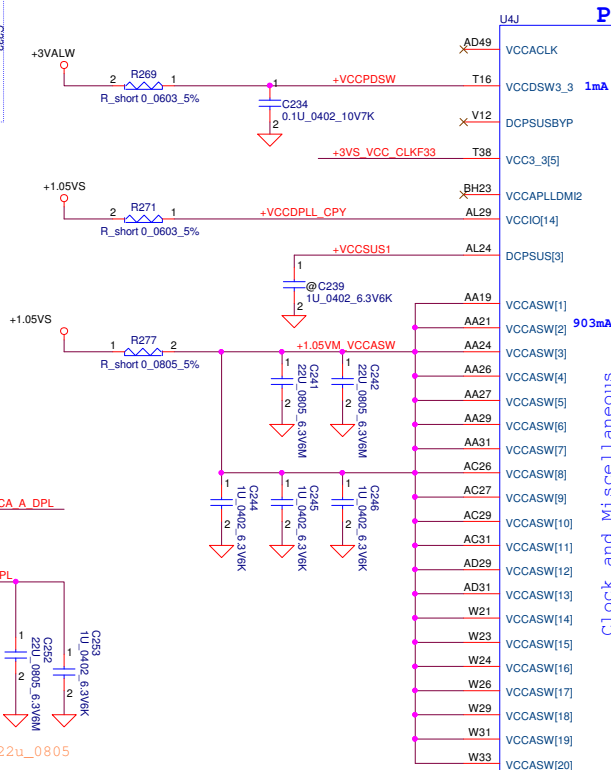
USB2 Port5, Charger Port

USB2 Port9, Right USB (Sub/B)



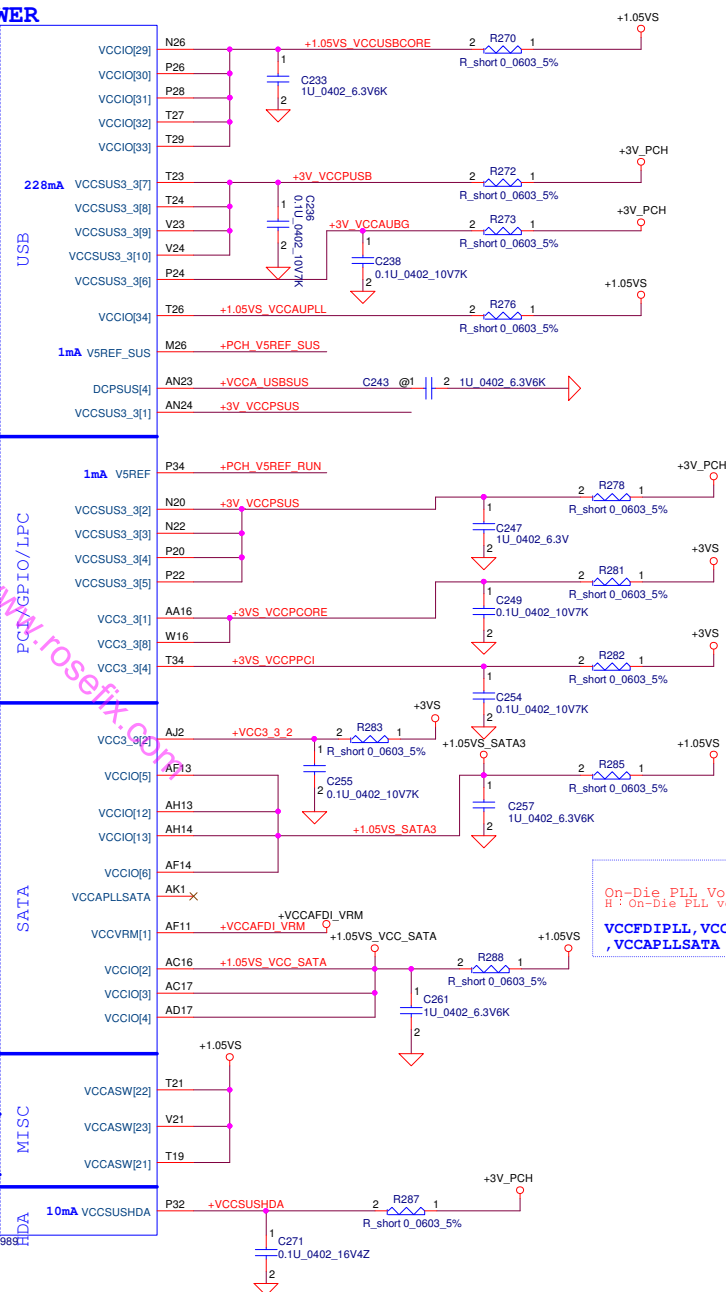


Have internal VRM

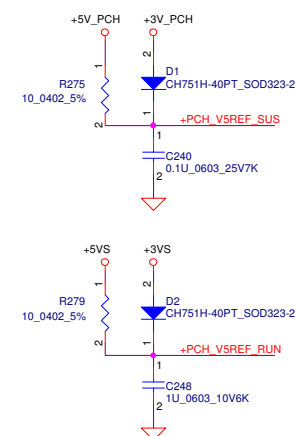


Before gerber out change to 22u_0805

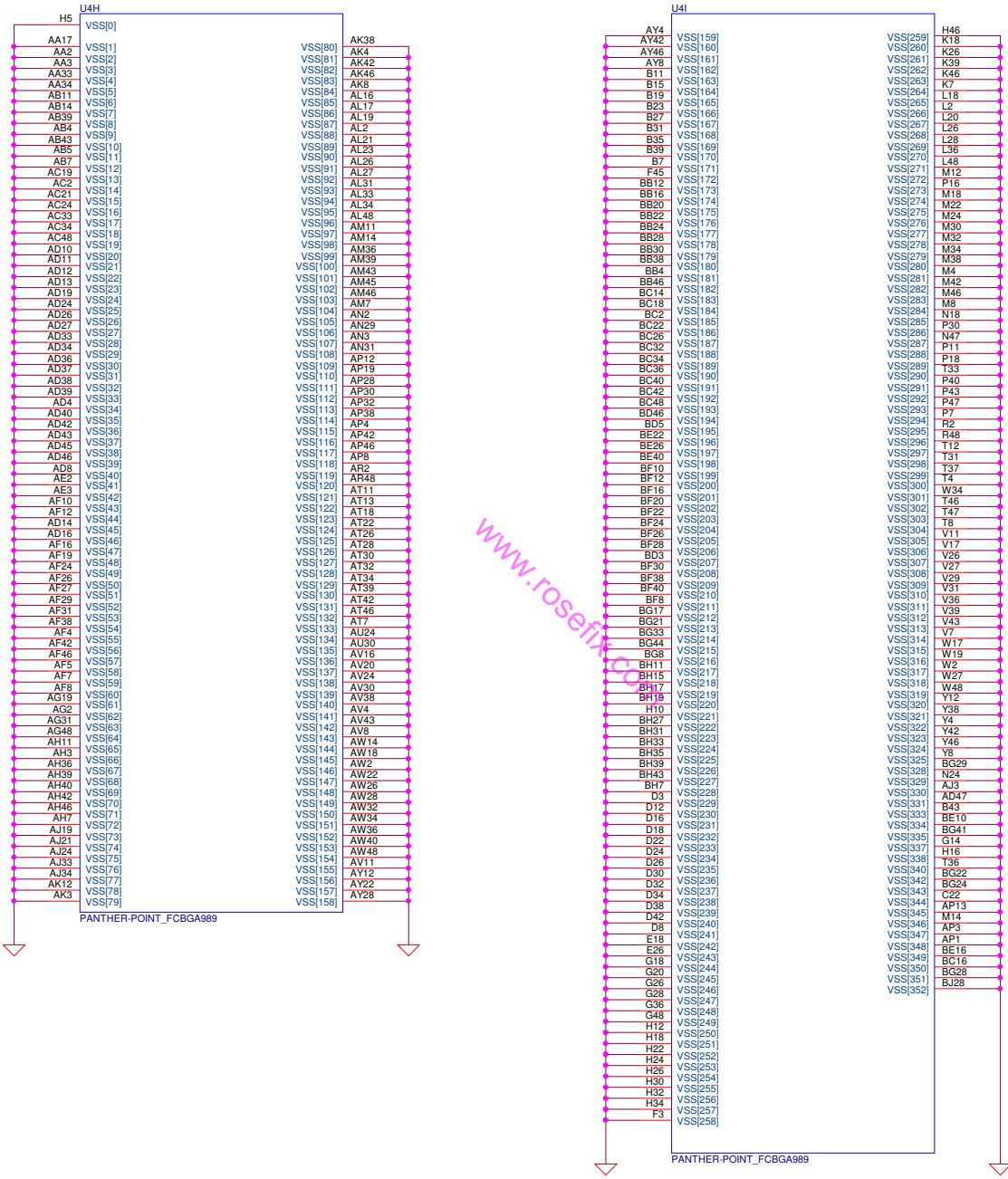
POWER



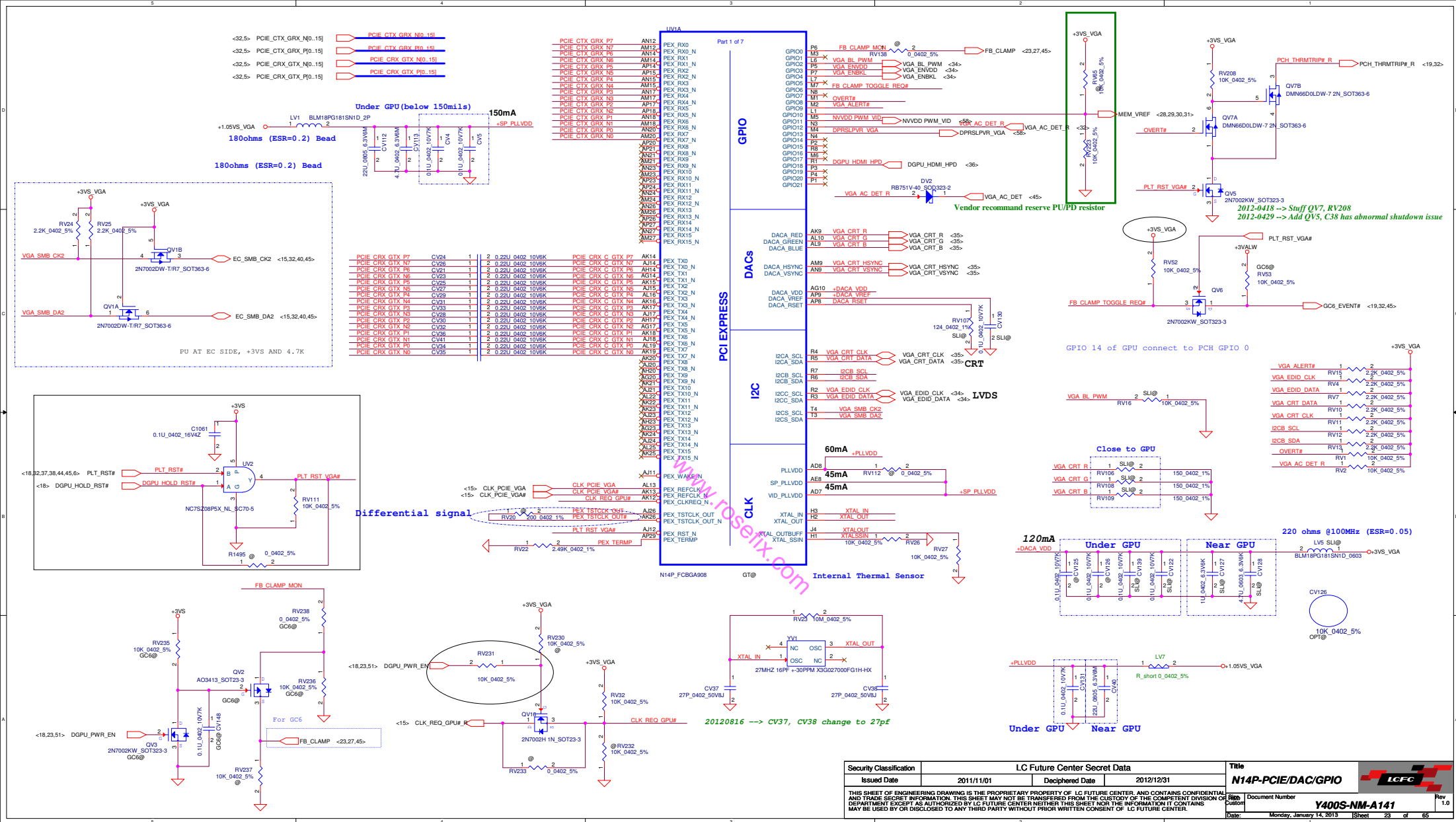
```
VCC3_3 = 266mA detal waiting for newest spec
VCCDMI = 42mA detal waiting for newest spec
```

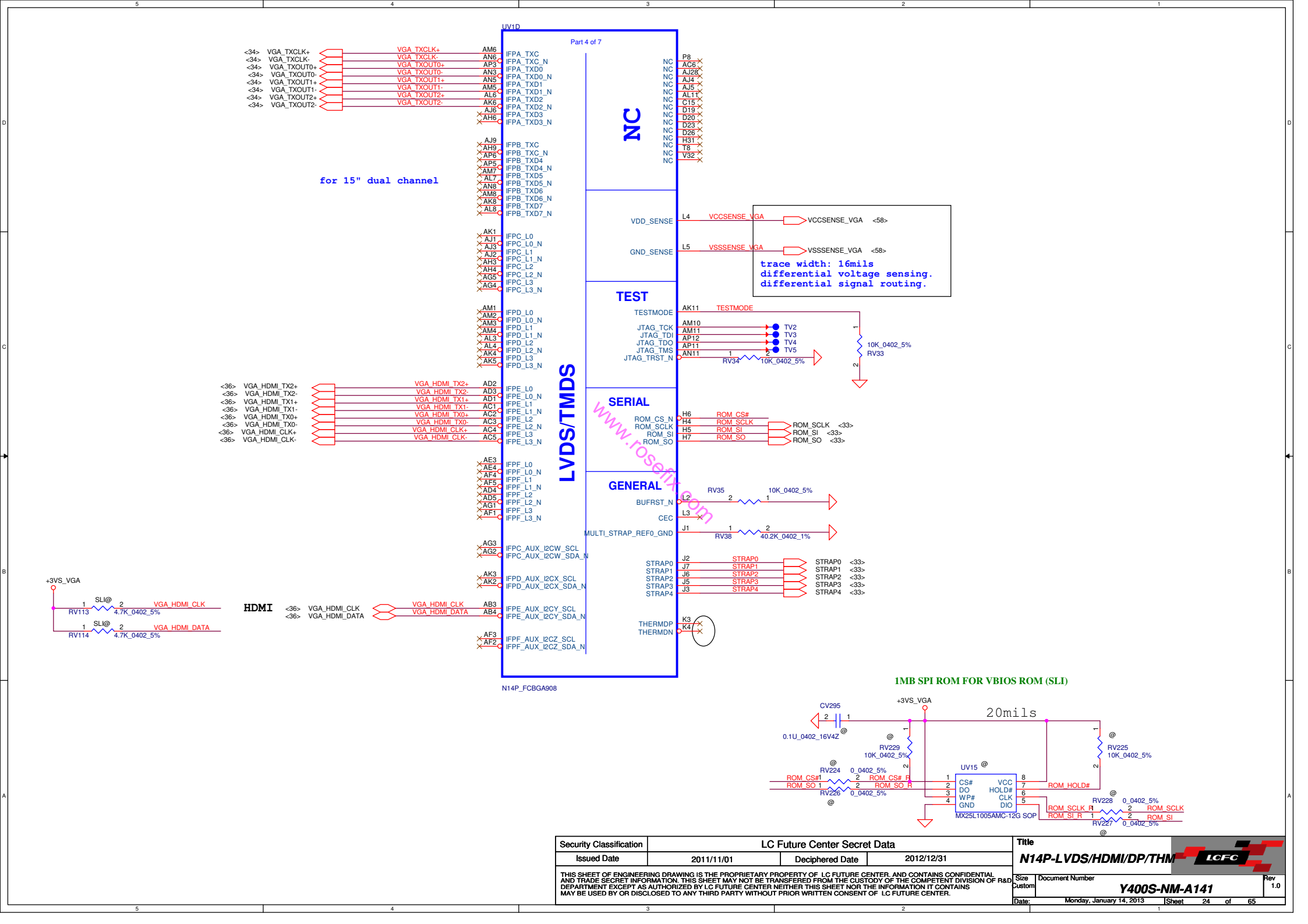


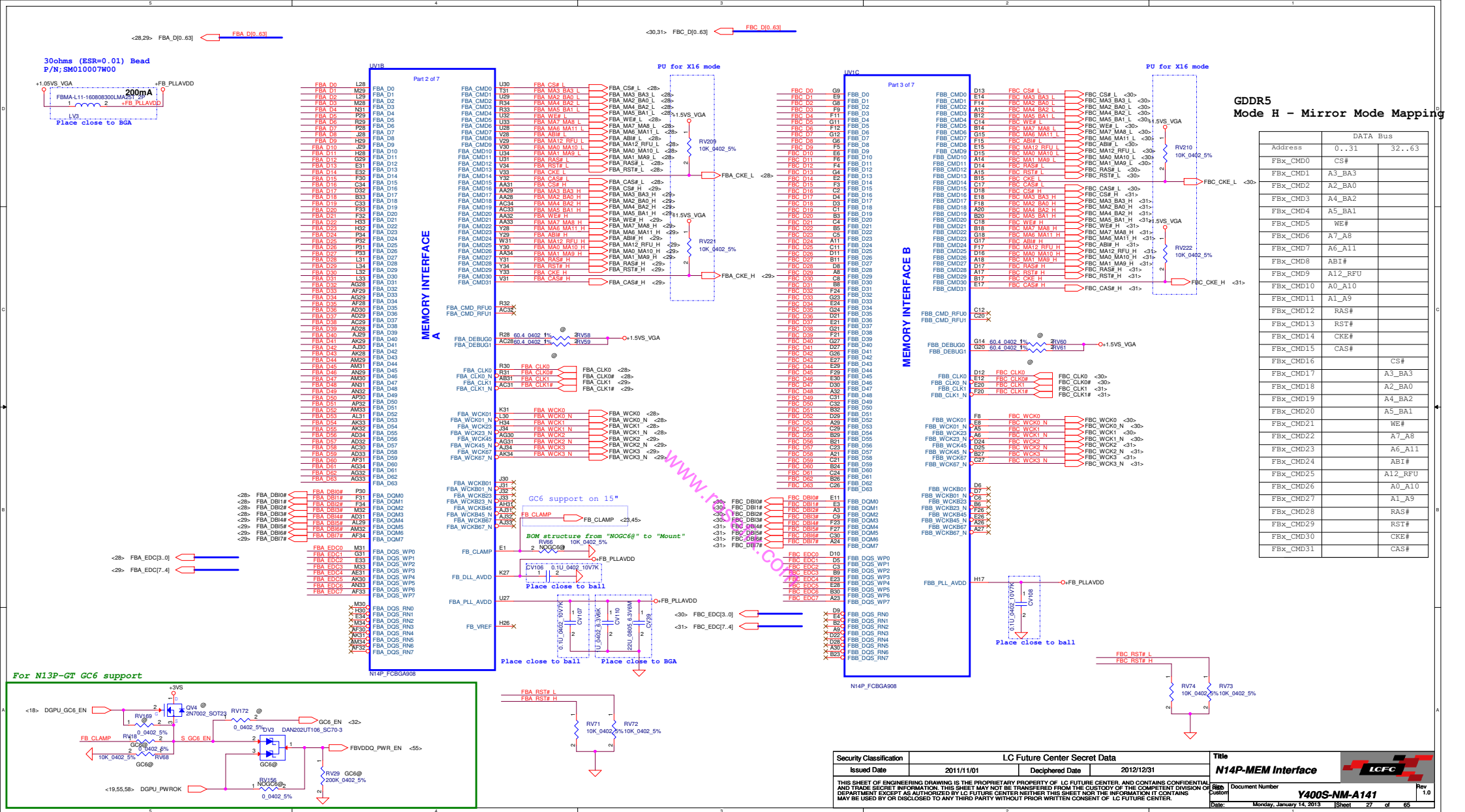
```
VCCFDIPLL, VCCAPLLEXP, VCCAPLLDMI2, VCCAPLLSATA
```



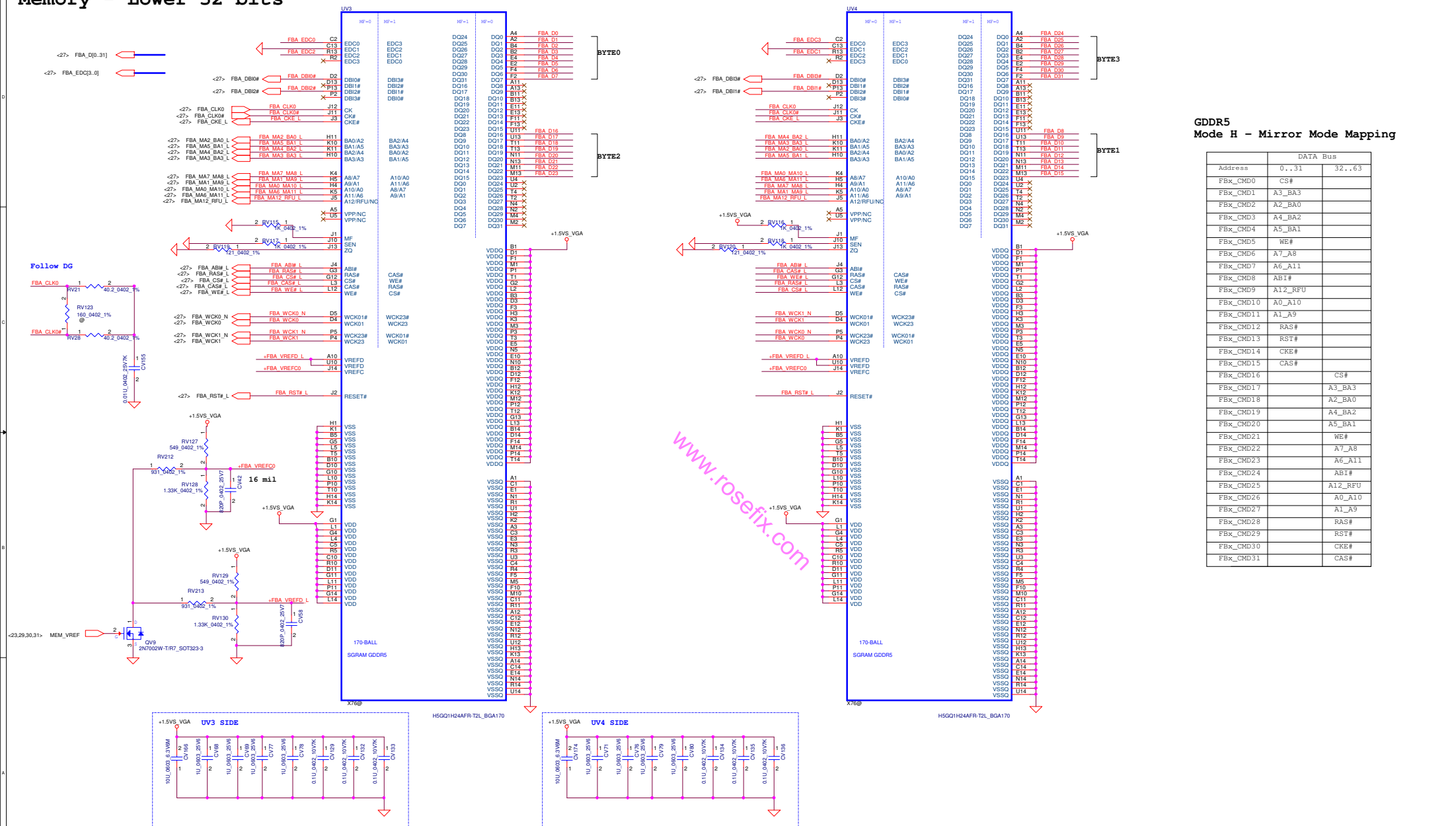
www.rosefix.com







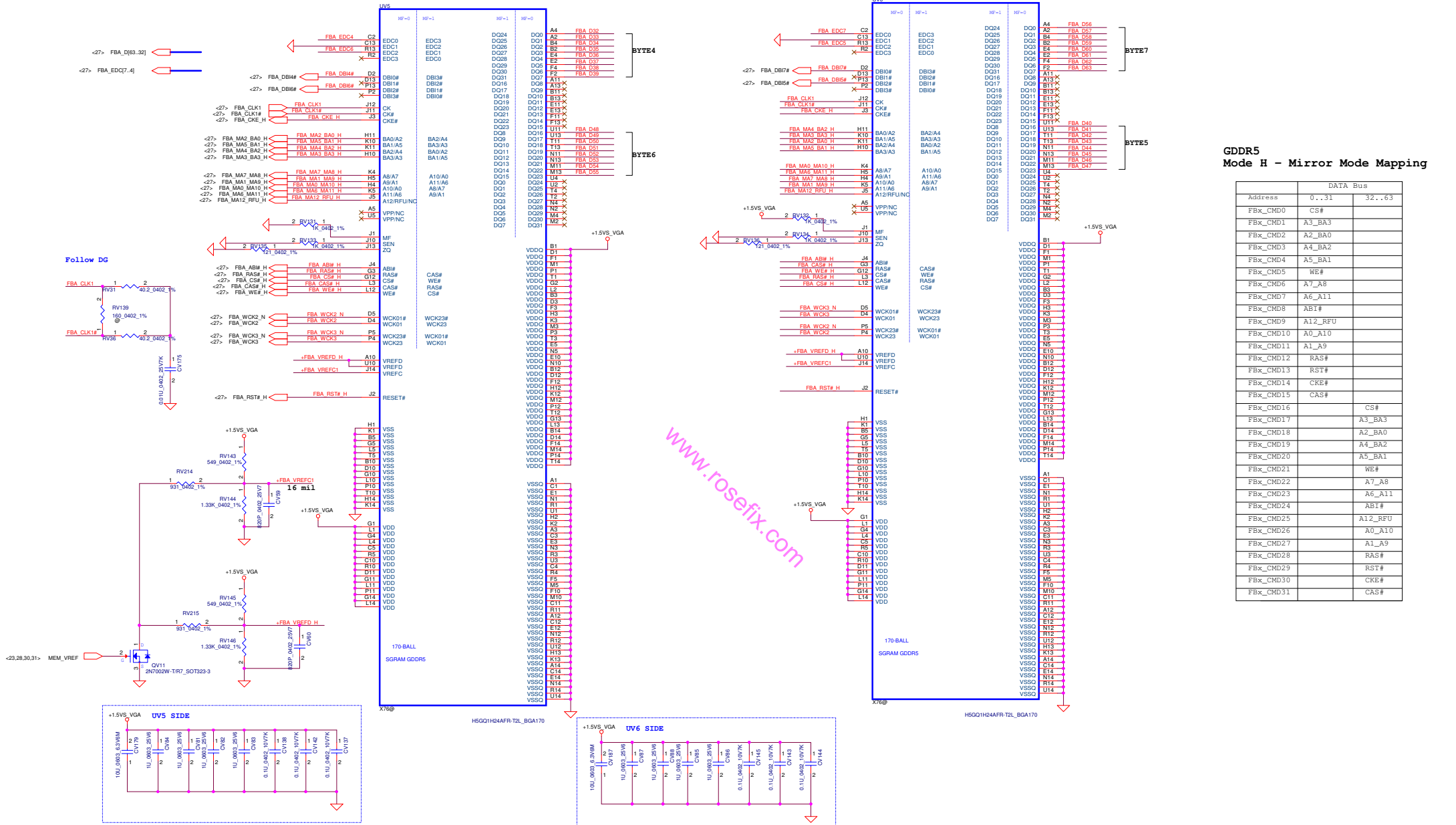
Memory - Lower 32 bits



GDDR5
Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FbX_CMD0	CS#	
FbX_CMD1	A3_BA3	
FbX_CMD2	A2_BA0	
FbX_CMD3	A4_BA2	
FbX_CMD4	A5_BA1	
FbX_CMD5	WE#	
FbX_CMD6	A7_A8	
FbX_CMD7	A6_A11	
FbX_CMD8	AB1#	
FbX_CMD9	A12_RFU	
FbX_CMD10	A0_A10	
FbX_CMD11	A1_A9	
FbX_CMD12	RA5#	
FbX_CMD13	RST#	
FbX_CMD14	CKE#	
FbX_CMD15	CAS#	
FbX_CMD16		CS#
FbX_CMD17		A3_BA3
FbX_CMD18		A2_BA0
FbX_CMD19		A4_BA2
FbX_CMD20		A5_BA1
FbX_CMD21		WE#
FbX_CMD22		A7_A8
FbX_CMD23		A5_A11
FbX_CMD24		AB1#
FbX_CMD25		A12_RFU
FbX_CMD26		A0_A10
FbX_CMD27		A1_A9
FbX_CMD28		RA5#
FbX_CMD29		RST#
FbX_CMD30		CKE#
FbX_CMD31		CAS#

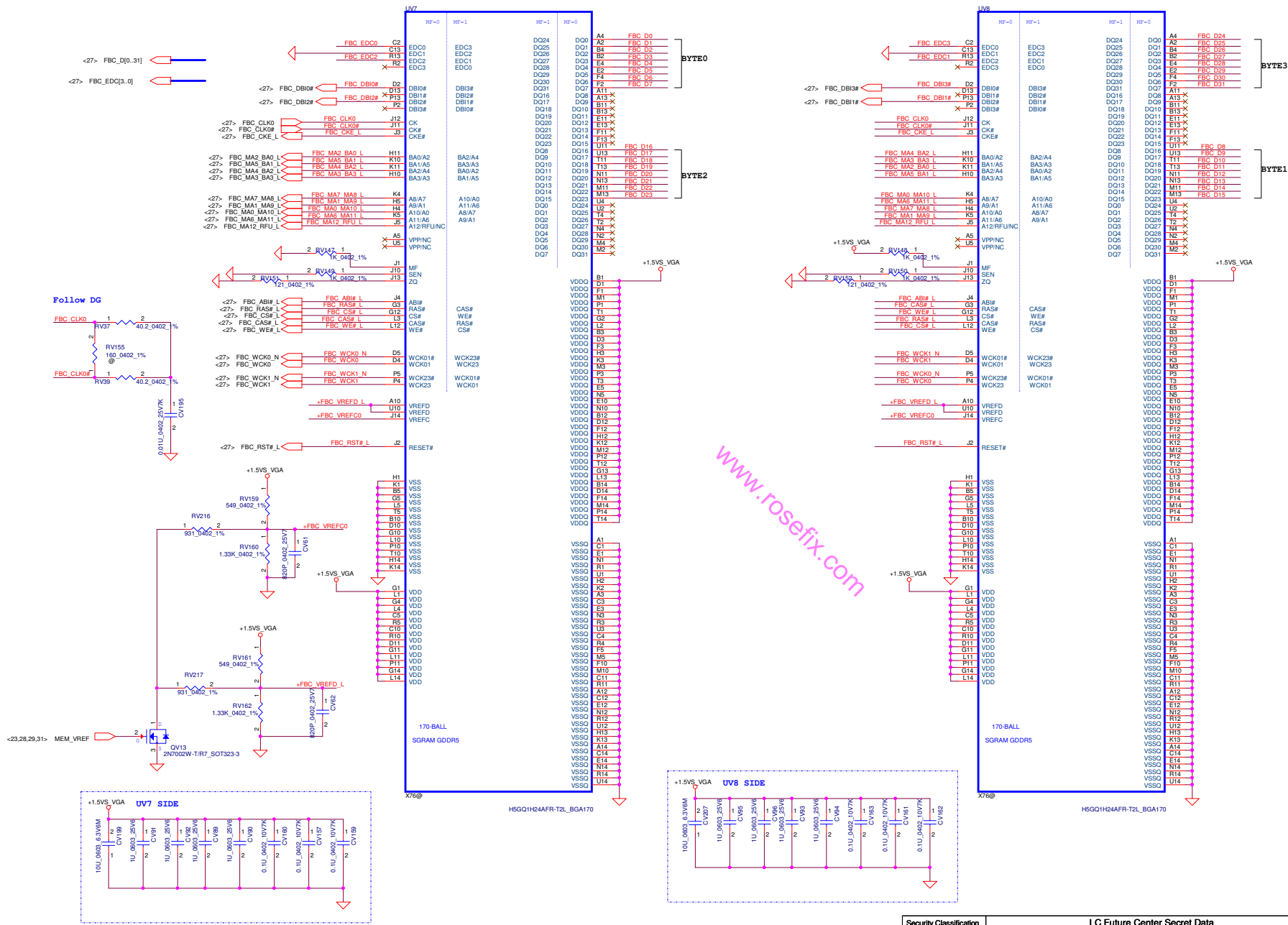
Memory - Upper 32 bits



GDDR5
Mode H - Mirror Mode Mapping

DATA Bus		
Address	0..31	32..63
Fbx_CMD0	CS#	
Fbx_CMD1	A3_BA3	
Fbx_CMD2	A2_BA0	
Fbx_CMD3	A2_BA2	
Fbx_CMD4	A5_BA1	
Fbx_CMD5	WE#	
Fbx_CMD6	A7_A8	
Fbx_CMD7	A6_A11	
Fbx_CMD8	AB1#	
Fbx_CMD9	A12_RFU	
Fbx_CMD10	A0_A10	
Fbx_CMD11	A1_A9	
Fbx_CMD12	RAS#	
Fbx_CMD13	RST#	
Fbx_CMD14	CKE#	
Fbx_CMD15	CAS#	
Fbx_CMD16		CS#
Fbx_CMD17	A3_BA3	
Fbx_CMD18	A2_BA0	
Fbx_CMD19	A2_BA2	
Fbx_CMD20	A5_BA1	
Fbx_CMD21	WE#	
Fbx_CMD22	A7_A8	
Fbx_CMD23	A6_A11	
Fbx_CMD24	AB1#	
Fbx_CMD25	A12_RFU	
Fbx_CMD26	A0_A10	
Fbx_CMD27	A1_A9	
Fbx_CMD28	RAS#	
Fbx_CMD29	RST#	
Fbx_CMD30	CKE#	
Fbx_CMD31	CAS#	

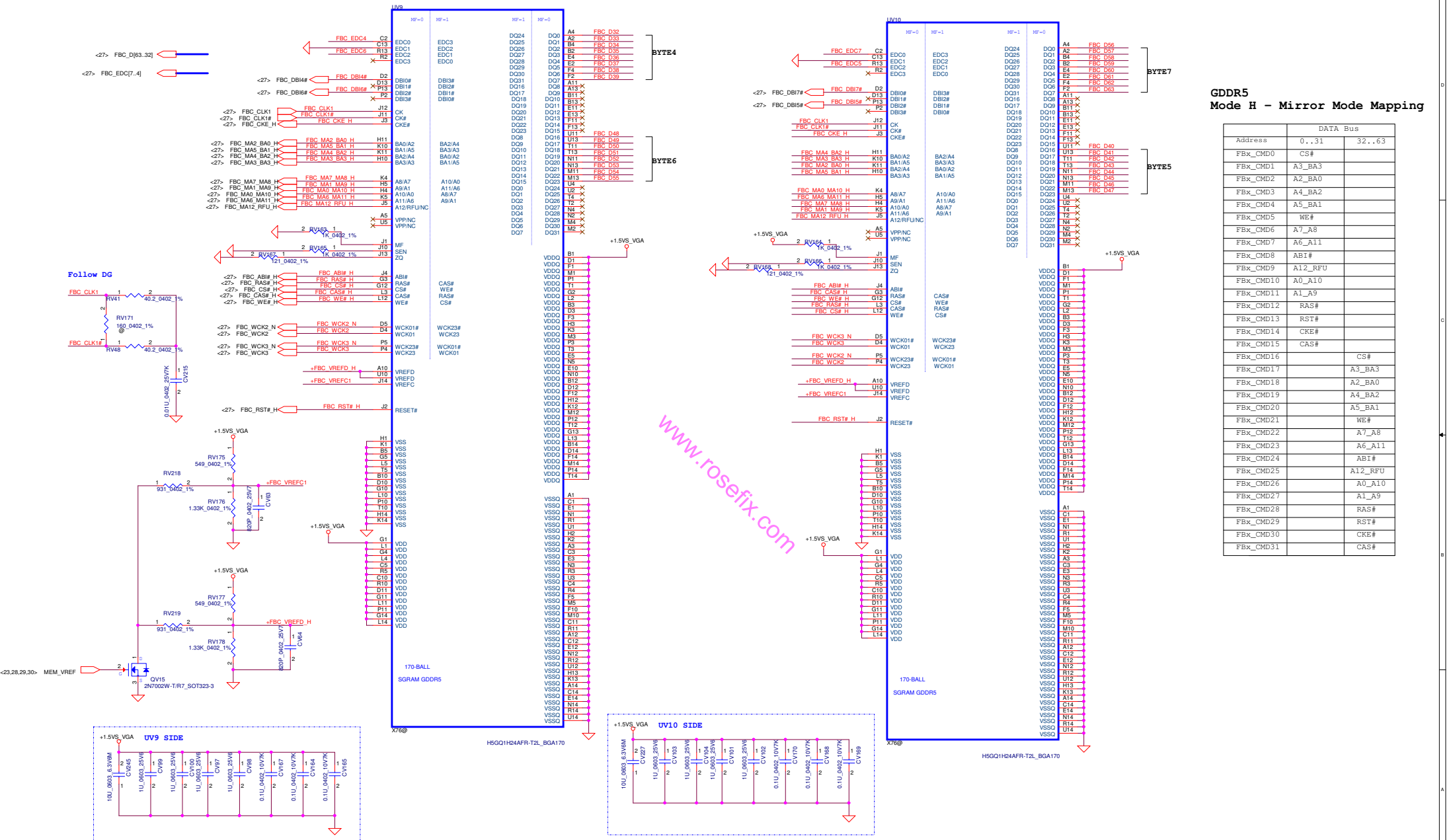
Memory Partition C - Lower 32 bits



GDDR5 Mode H - Mirror Mode Mapping

DATA Bus	
Address	0..31 32..63
FBX_CMD0	CS#
FBX_CMD1	A3_BA3
FBX_CMD2	A2_BA0
FBX_CMD3	A4_BA2
FBX_CMD4	A5_BA1
FBX_CMD5	WE#
FBX_CMD6	A7_A8
FBX_CMD7	A6_A11
FBX_CMD8	AB1#
FBX_CMD9	A12_RFU
FBX_CMD10	A0_A10
FBX_CMD11	A1_A9
FBX_CMD12	RAS#
FBX_CMD13	RST#
FBX_CMD14	CKE#
FBX_CMD15	CAS#
FBX_CMD16	
FBX_CMD17	A3_BA3
FBX_CMD18	A2_BA0
FBX_CMD19	A4_BA2
FBX_CMD20	A5_BA1
FBX_CMD21	WE#
FBX_CMD22	A7_A8
FBX_CMD23	A6_A11
FBX_CMD24	AB1#
FBX_CMD25	A12_RFU
FBX_CMD26	A0_A10
FBX_CMD27	A1_A9
FBX_CMD28	RAS#
FBX_CMD29	RST#
FBX_CMD30	CKE#
FBX_CMD31	CAS#

Memory Partition C - Upper 32 bits

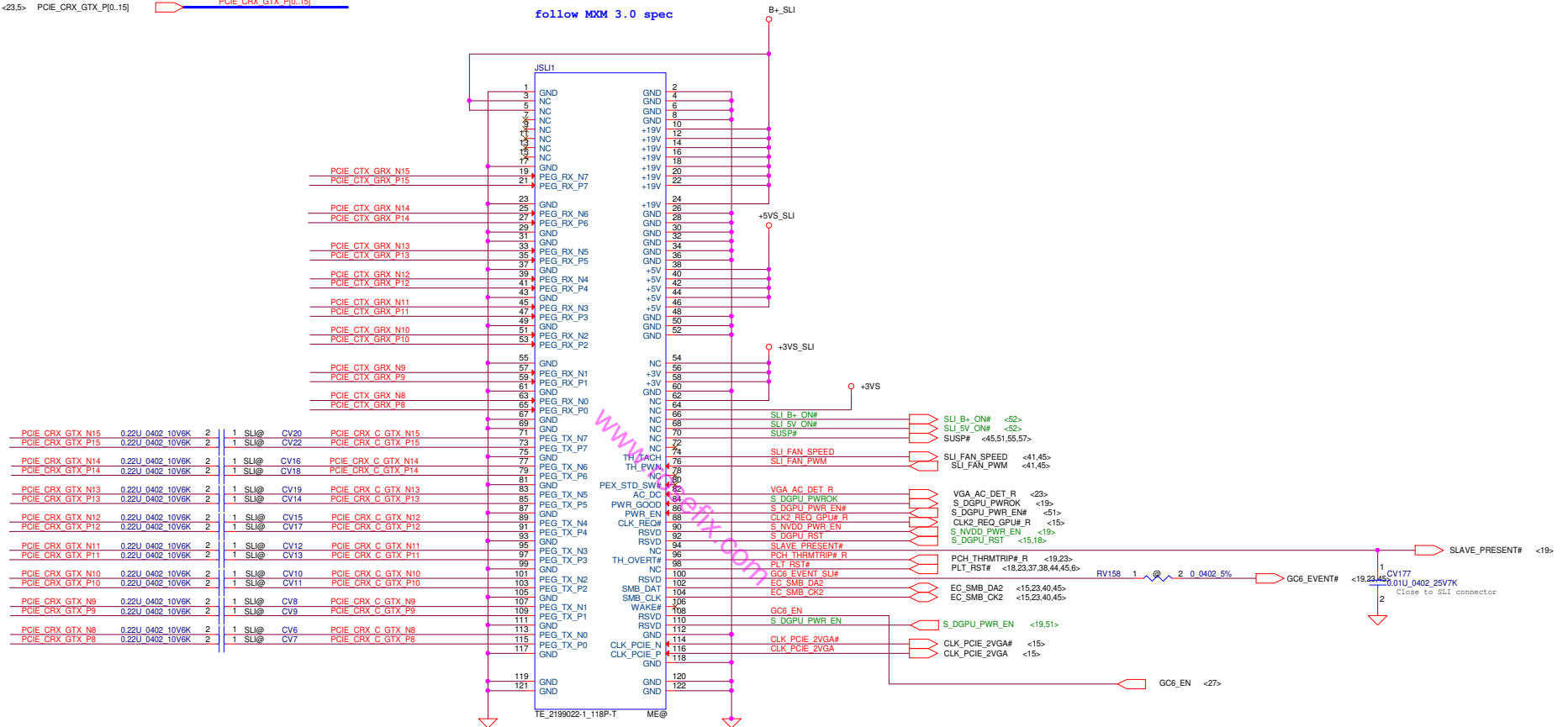


11/11 for 2nd VGA fan
need to notice EC

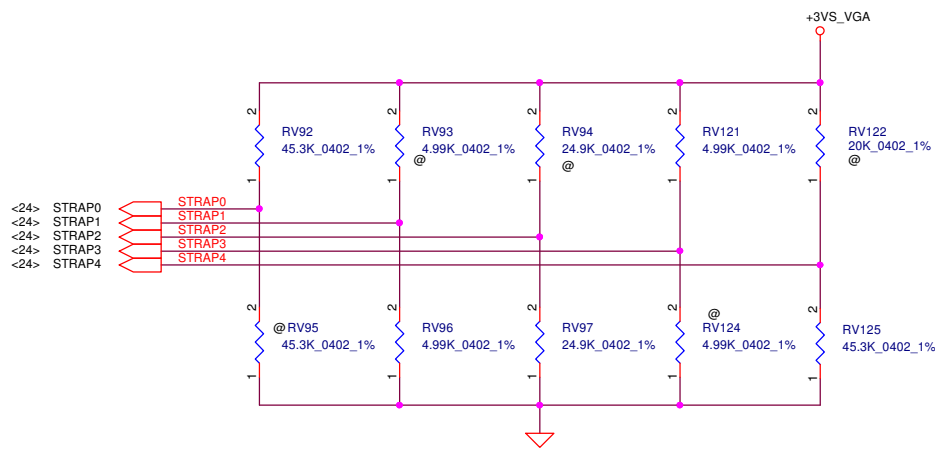
<23.5> PCIE_CTX_GRX_N[0..15]
<23.5> PCIE_CTX_GRX_P[0..15]
<23.5> PCIE_CRX_GTX_N[0..15]
<23.5> PCIE_CRX_GTX_P[0..15]

PCIE_CTX_GRX_N[0..15]
PCIE_CTX_GRX_P[0..15]
PCIE_CRX_GTX_N[0..15]
PCIE_CRX_GTX_P[0..15]

follow MXM 3.0 spec



Security Classification		LC Future Center Secret Data		Title	
Issued Date		2011/11/01		2ND VGA CONN.	
Deciphered Date		2012/12/31		Size	
Document Number		Y400S-NM-A141		Rev	
Date		Monday, March 11, 2013		Sheet	
32		of		85	



Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

Resistor Values	Pull-up to +3VS_VGA	Pull-down to Gnd
5K	1000	0000
10K	1001	0001
15K	1010	0010
20K	1011	0011
25K	1100	0100
30K	1101	0101
35K	1110	0110
45K	1111	0111

SLOT_CLK_CFG	
0	GPU and MCH don't share a common reference clock
1	GPU and MCH share a common reference clock (Default)

SUB_VENDOR	
0	No VBIOS ROM (Default)
1	BIOS ROM is present

3GIO_PADCFG	
3GIO_PADCFG[3:0]	
0000	Notebook Default

XCLK_417	
0	277MHz (Default)
1	Reserved

USER Straps	
User[3:0]	
1000-1100	Customer defined

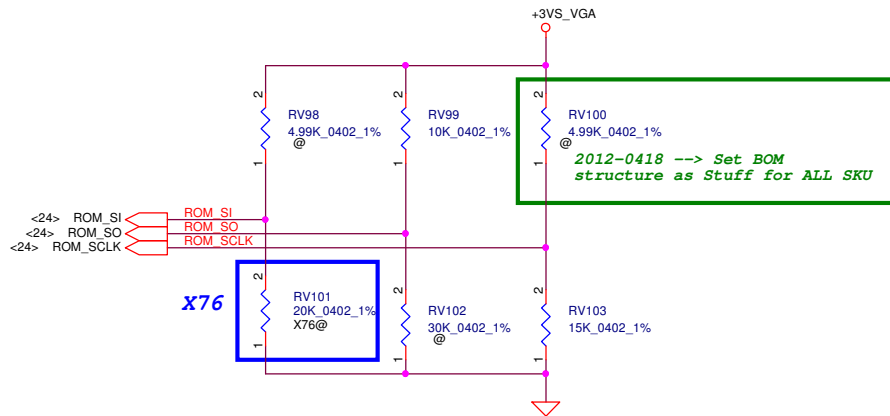
PEX PLL EN TERM	
0	Disable (Default)
1	Enable

PCIE_MAX_SPEED	
0	Limit to PCIe Gen1
1	PCIe Gen 2/3 Capable

FB_0_BAR_SIZE	
0	Reserved
1	Reserved
2	256MB (Default)
3	Reserved

SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

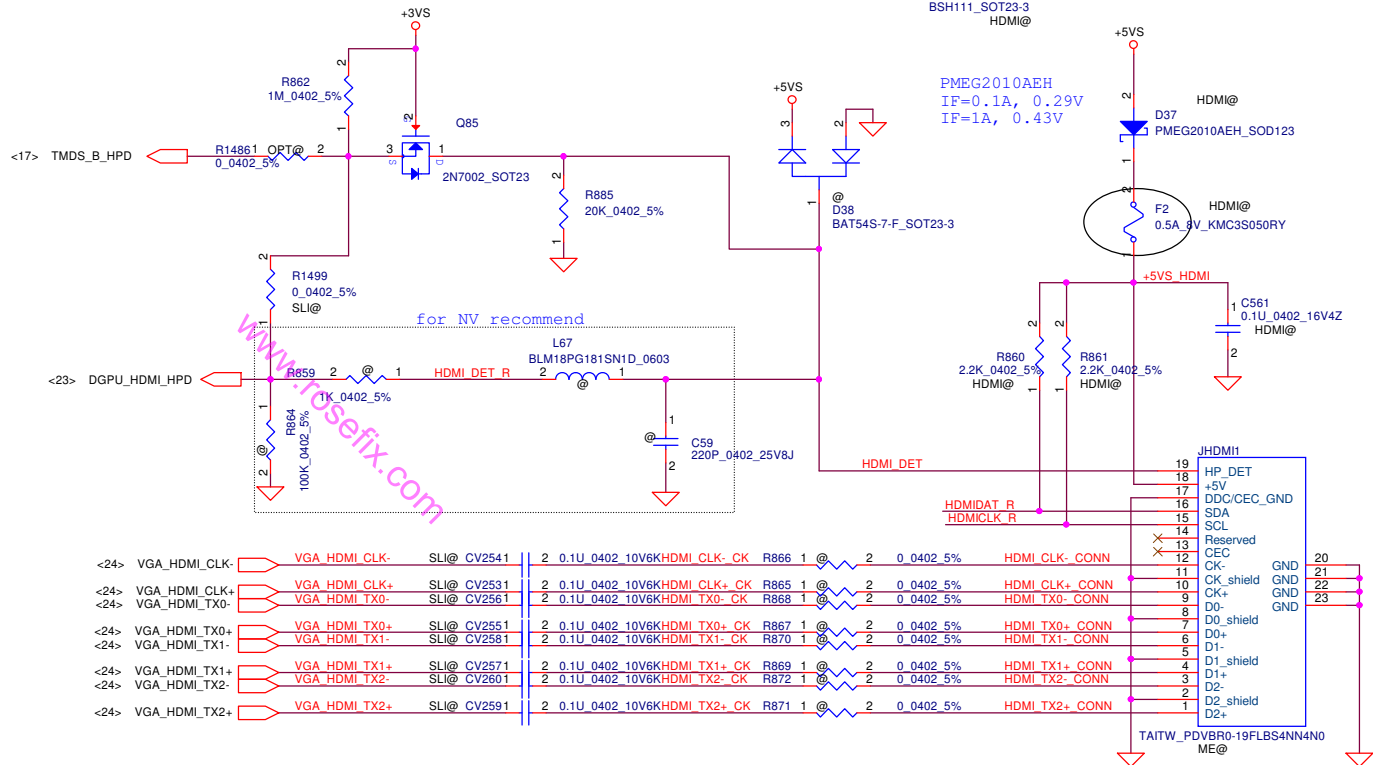
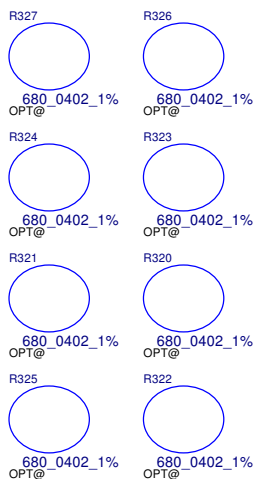
VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)



GPU	FB Memory (GDDR5)		ROM_SI	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4
N13P-GT1 28nm	Samsung	K4G20325FD-FC04 2G 64Mx32	PD 30K	PU 10K	PU 35K (ALL SKU)	PU 45K	PD 5K	PD 25K	PU 5K	PD 45K
		K4G10325FG-HC04 1G 32Mx32	PD 45K							
	Hynix	H5GQ2H24MFR-T2C 2G 64Mx32	PD 25K							
		H5GQ1H24BFR-T2C 1G 32Mx32	PD 20K							
		H5GQ2H24AFR-T2C 2G 64Mx32	PD 5K							

VRAM	X76	VRAM P/N
Samsung	X76409JVL01 (2G 64Mx32)	SA00005B70J
	X76409JVL51 (1G 32Mx16)	SA00003RS0J
Hynix	X76409JVL02 (2G 64Mx32)	SA00004GD0J EOL
	X76409JVL02 (2G 64Mx32)	SA00004GD1J
	X76409JVL52 (1G 32Mx16)	SA00003WL1J

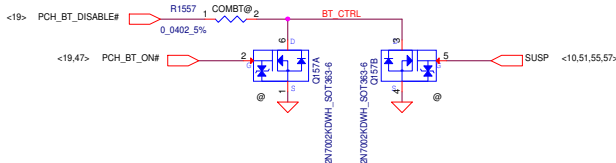
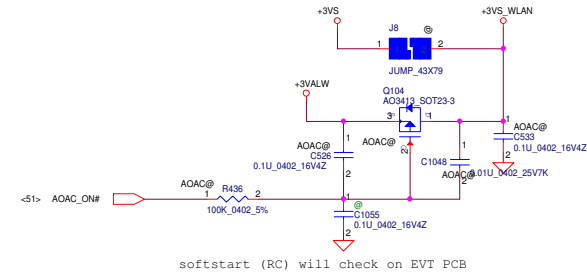
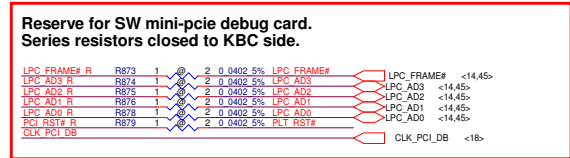
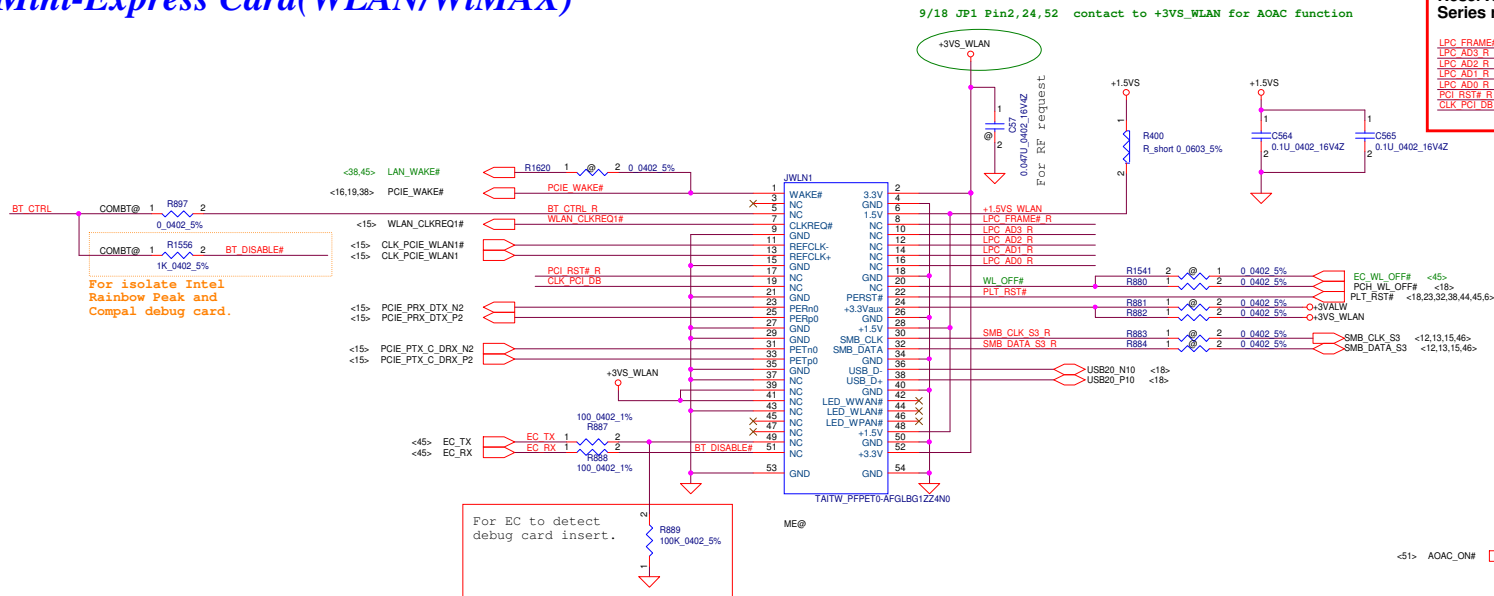
Security Classification		LC Future Center Secret Data		Title	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	N14P_MISC	
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				Date: Monday, January 14, 2013	Sheet 33 of 65



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Security Classification		LC Future Center Secret Data		Title			
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				Custom	Y400S-NM-A141	1.0	
				Date:	Monday, March 11, 2013	Sheet	36 of 65

Mini-Express Card(WLAN/WiMAX)

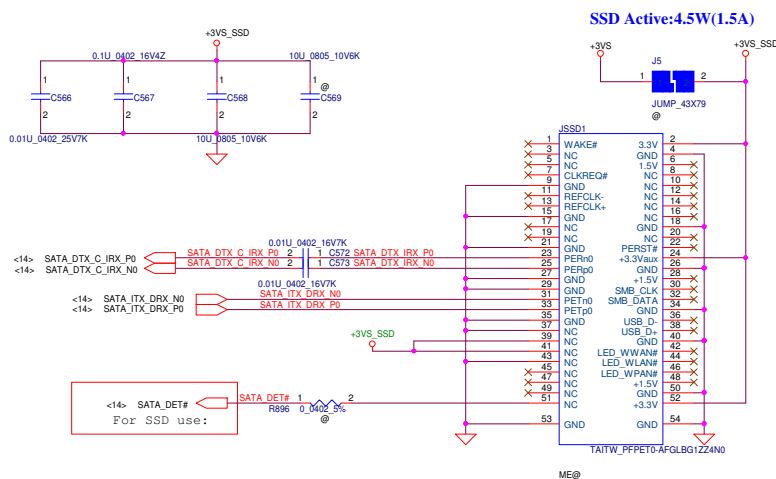


WLAN&BT Combo module circuits

	BT on module Enable	BT on module Disable
* BT_CTRL (GPIO22)	H	L
PCH_BT_ON#	L	H

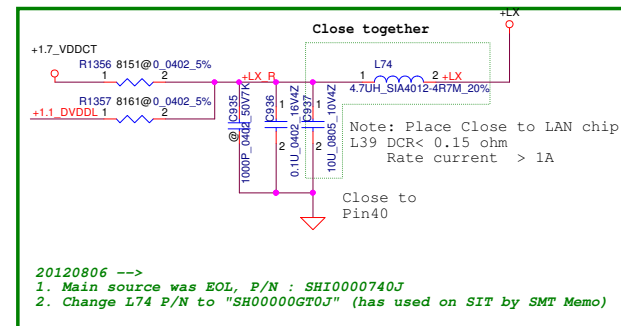
9/18 Increase for Intel AOAC function

Mini-Express Card(SSD)



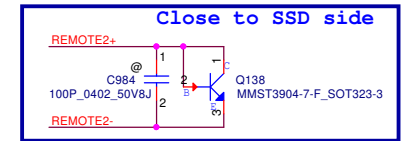
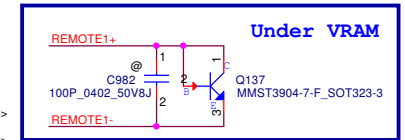
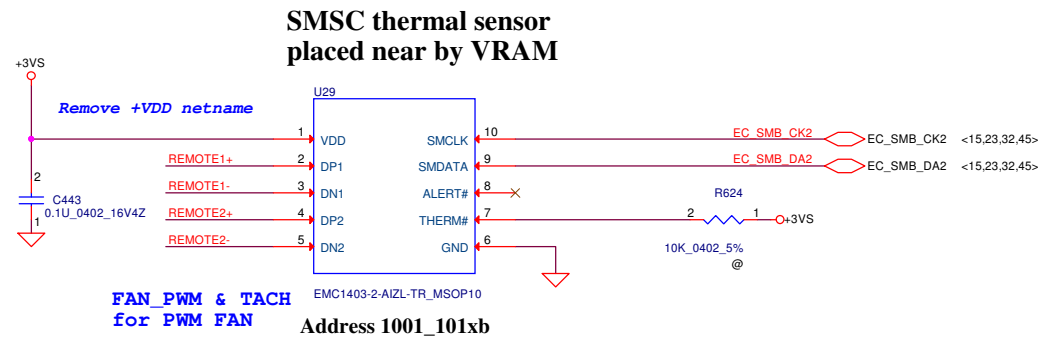
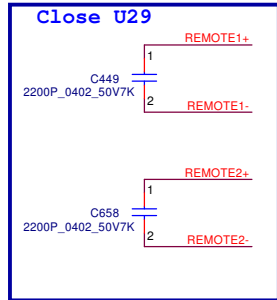
Security Classification	LC Future Center Secret Data		Title	MINI-CARD CONN.	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	Size	Document Number
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Date	Monday, March 11, 2013	Sheet	37	of	65

Layout Notice : Place as close chip as possible.

[illegible]

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Issued Date	2011/11/01	Deciphered Date	2012/12/31
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Title		LAN-AR8151/8161			
R & S	Size Custom	Document Number			Rev 1.0
		Y400S-NM-A141			
	Date:	Monday, March 11, 2013	Sheet	38 of 65	

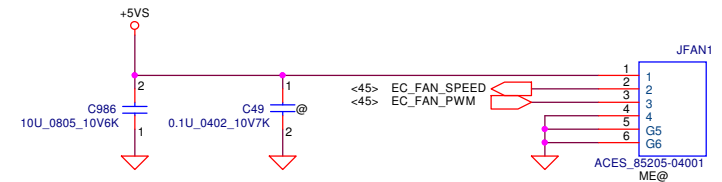


internal pull up 1.2K to 1.5V
R for initial thermal
shutdown temp

REMOTE2+/-:
Trace width/space:10/10 mil
Trace length:<8"

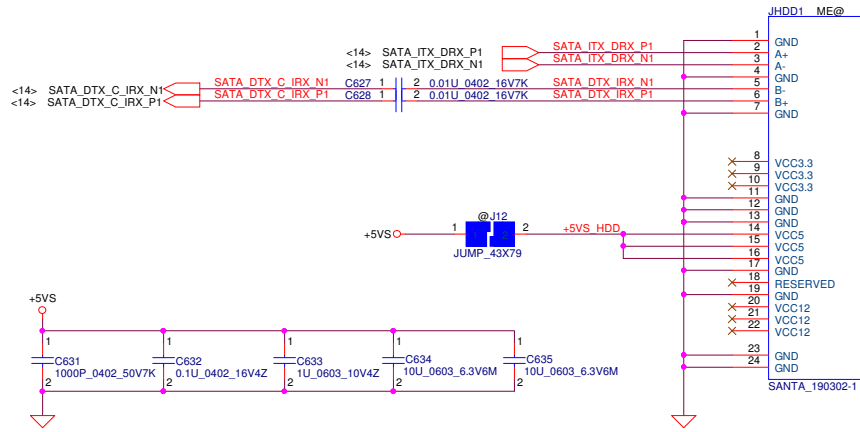
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FAN1 Conn

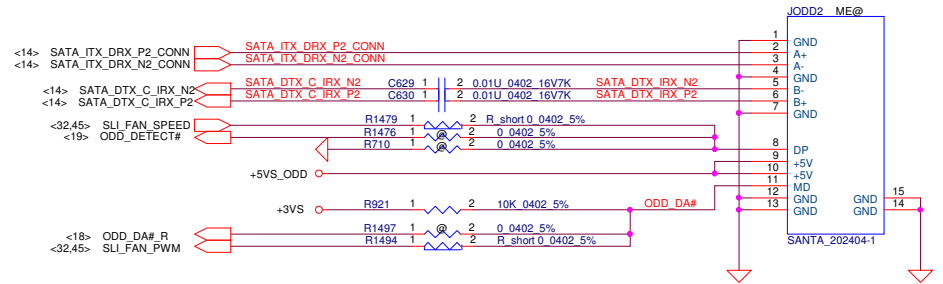


Security Classification		LC Future Center Secret Data		Title	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	EMC THERMAL SENSOR/FAN	
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				Y400S-NM-A141	
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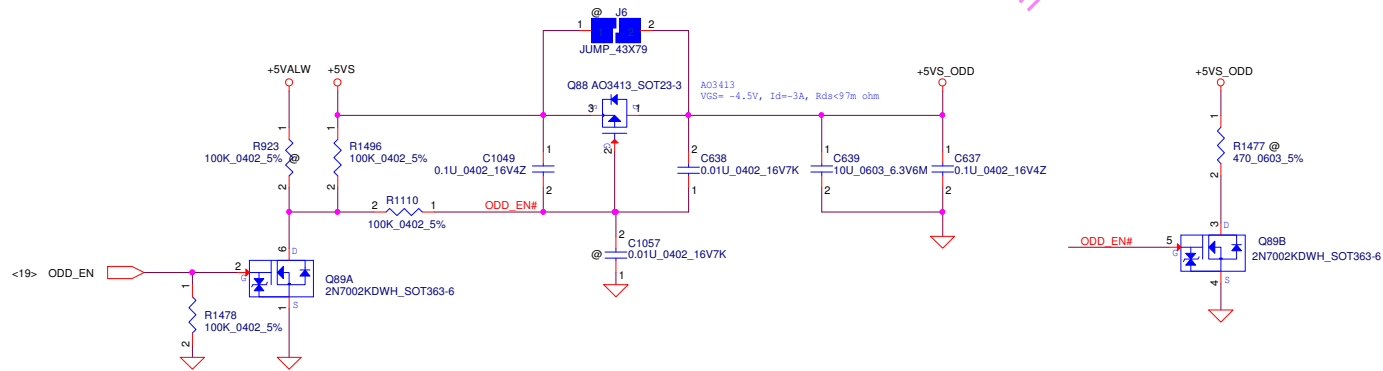
SATA HDD Conn.

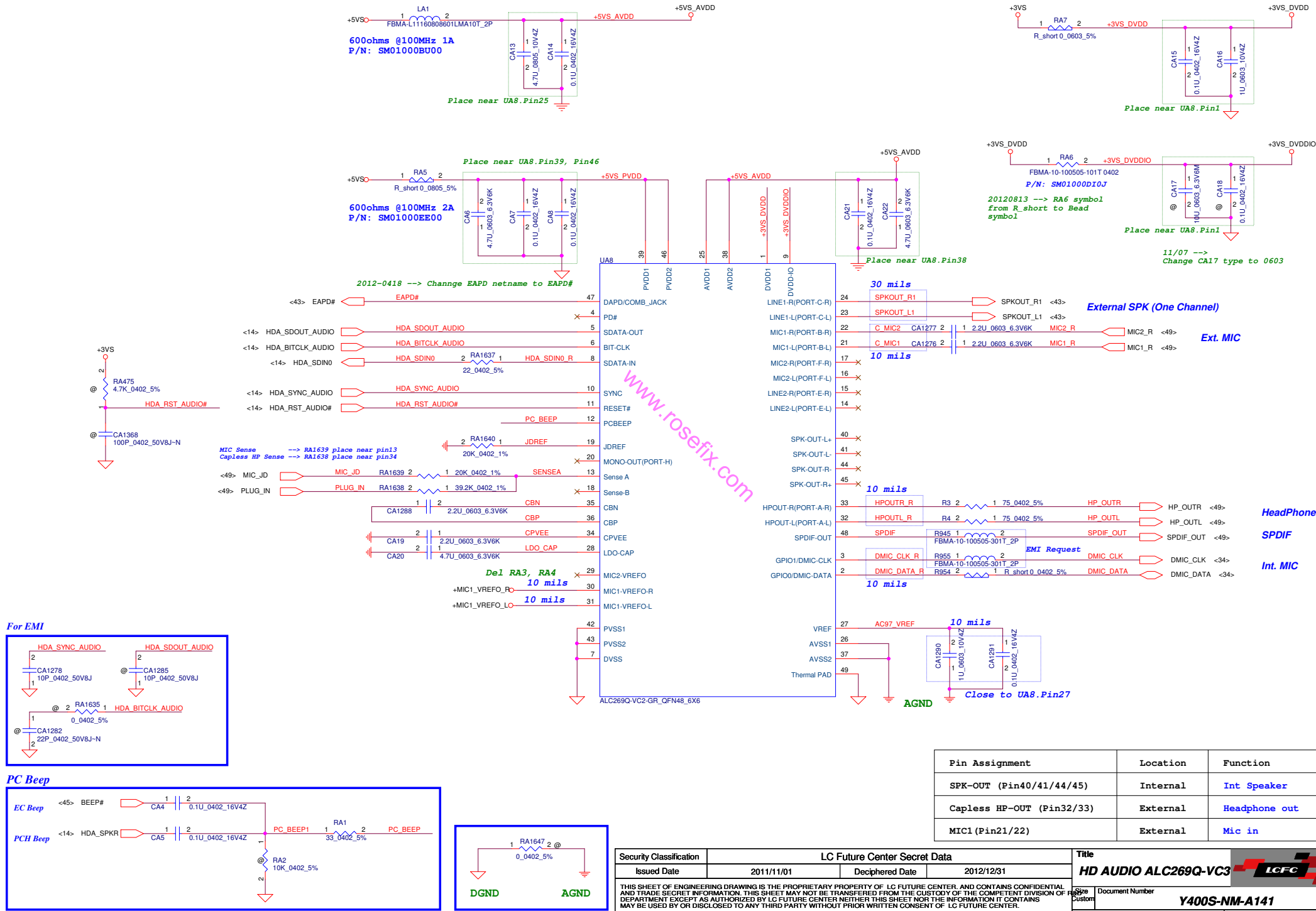


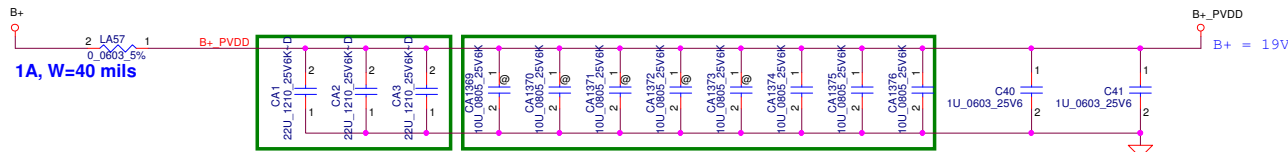
SATA ODD Conn.



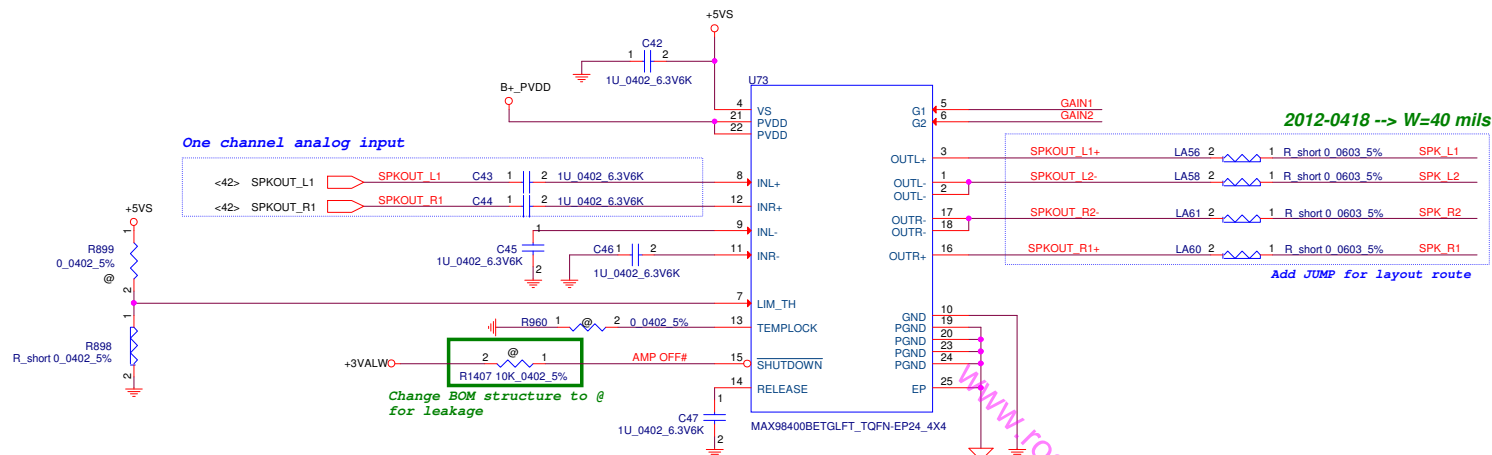
ODD Power Control







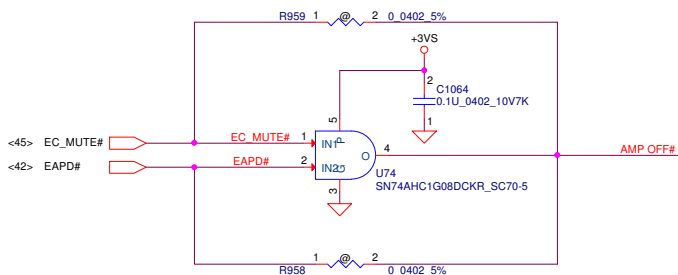
22uF*3, 10uF*8 for Damage issue
Stuff --> CA1, CA2, CA3, CA1374, CA1375 and CA1376



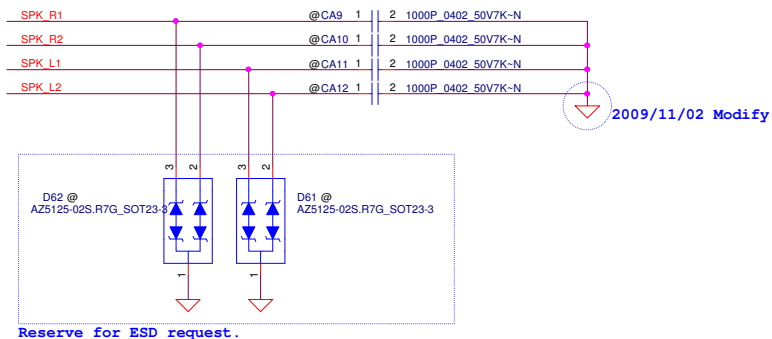
2012-0418 --> W=40 mils

Add JUMP for layout route

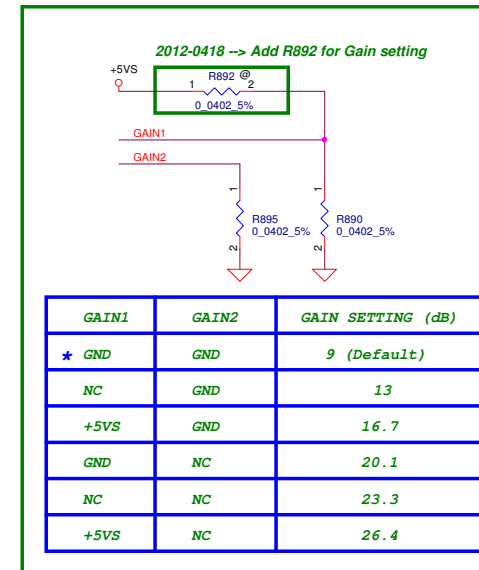
2012-0429 --> Change C42-C47 Cap to X5R type for Vendor suggestion



2012-0622 --> change these BOM structure for BOBO noise issue
1. R958, R959 --> "UnStuff"
2. U74, C1064 --> "Stuff"



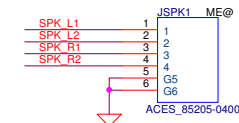
GAIN SETTING

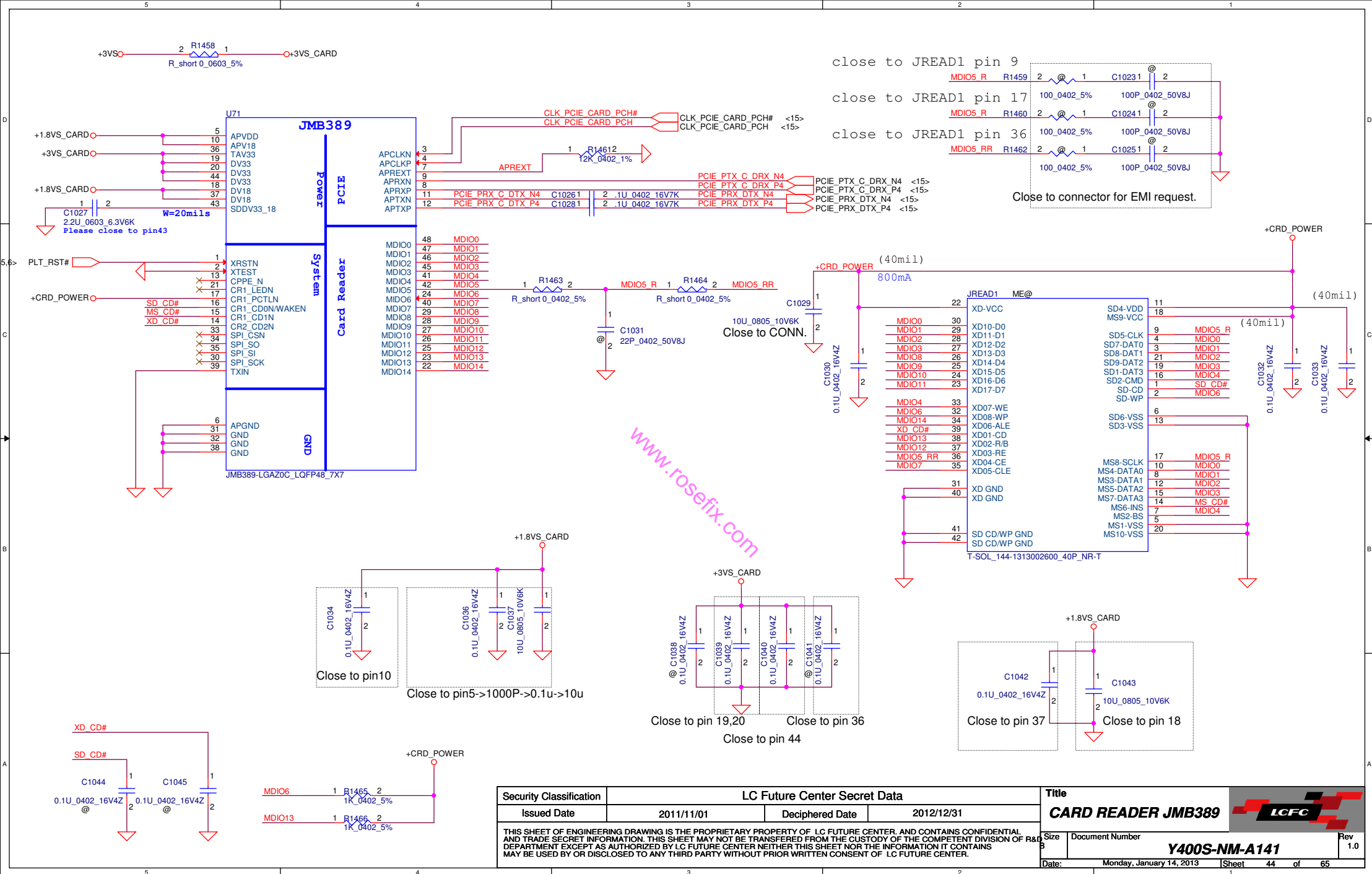


2012-0418 --> Set R890 BOM structure as Stuff

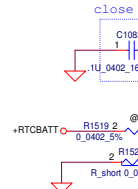
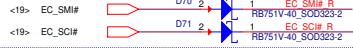
GAIN1	GAIN2	GAIN SETTING (dB)
* GND	GND	9 (Default)
NC	GND	13
+5VS	GND	16.7
GND	NC	20.1
NC	NC	23.3
+5VS	NC	26.4

Speaker Conn.

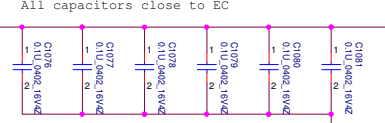




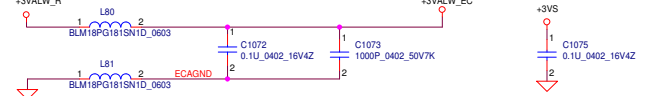
EC_SCI# / EC_SMI# pull up to PCH



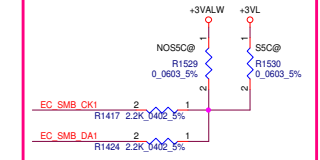
Support DC S5 Charge



minimum trace width 12 mil

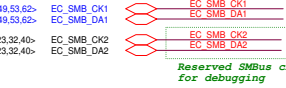
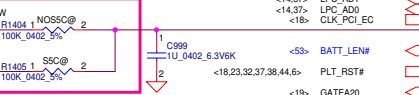


Support DC S5 Charge

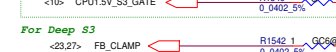
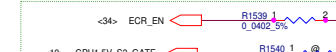


20120713 --> change to normal footprint

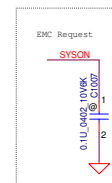
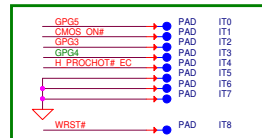
Support DC S5 Charge



Please place R1435 close to EC within 790mil

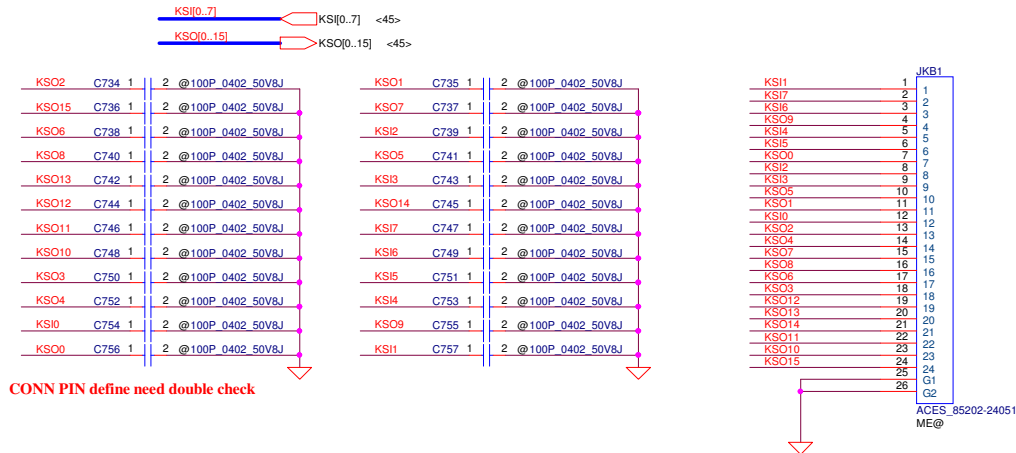


For factory EC flash

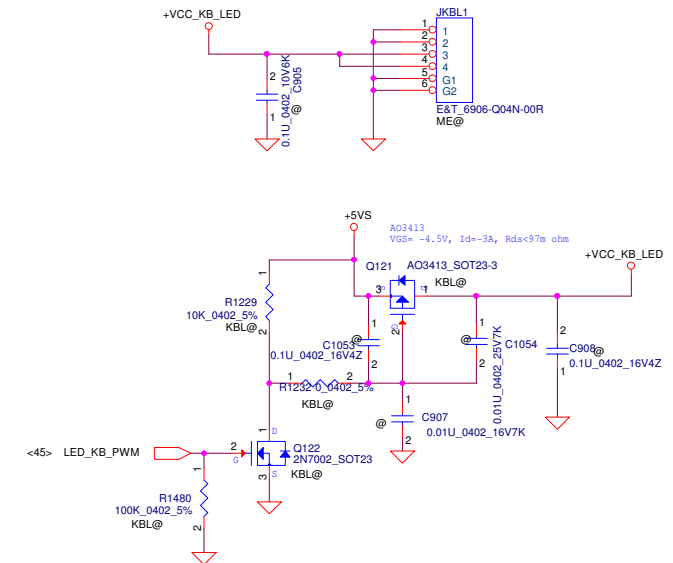


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2011/11/01		2012/12/31			
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				Y400S-NM-A141	
				Date:	Monday, January 14, 2013
				Sheet	45 of 65
				Rev	1.0

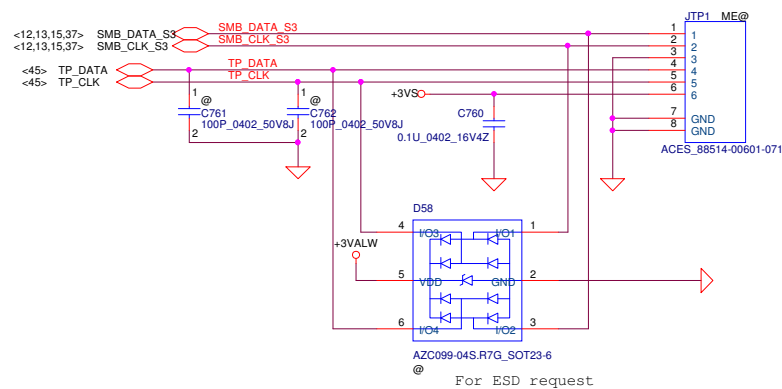
14" INT_KBD Conn.



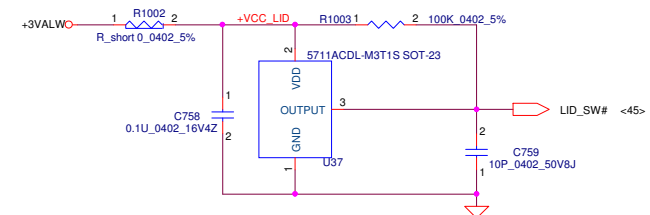
KB Lighting CONN.4pin



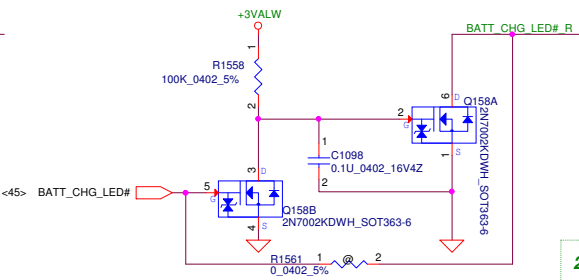
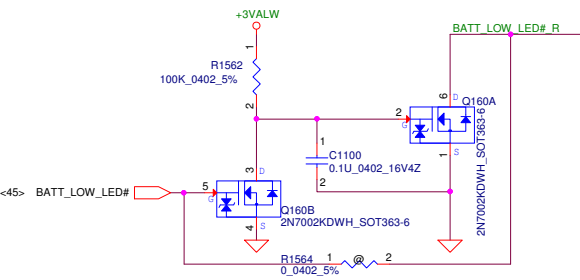
To TP/B Conn.



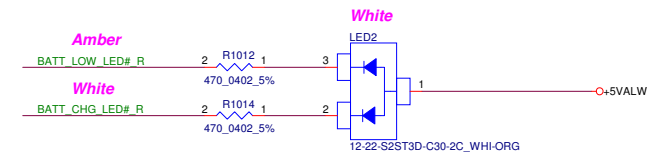
Lid Switch



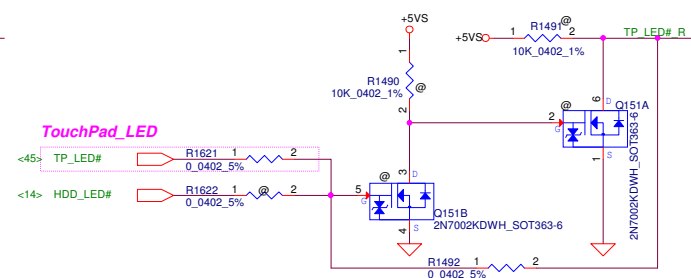
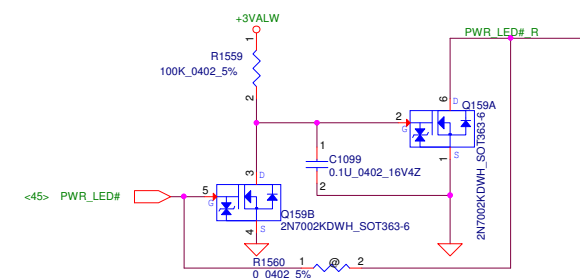
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Issued Date	2011/11/01	Deciphered Date	2012/12/31	Document Number	Y400S-NM-A141
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				Date:	Monday, January 14, 2013
				Sheet	46 of 65



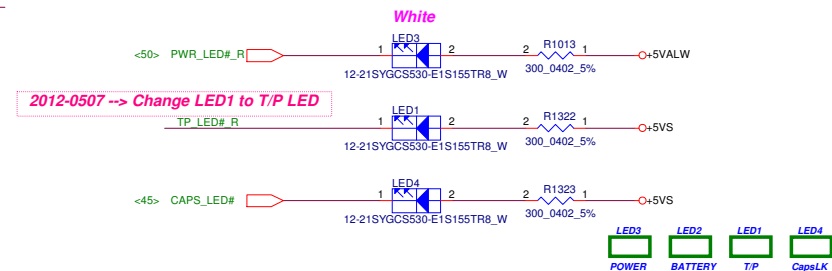
BATT CHARGE/LOW LED



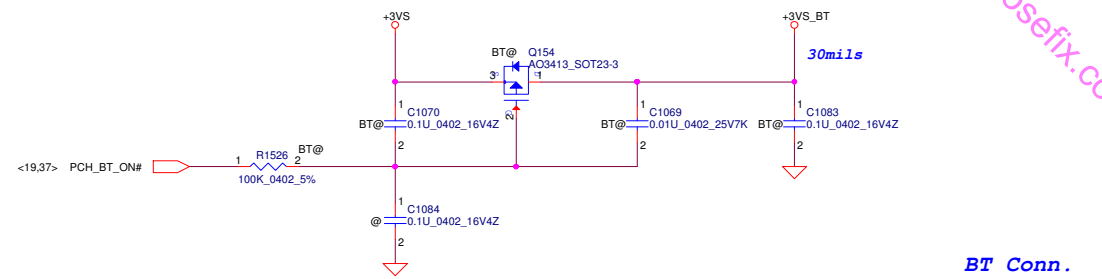
2012-0507 --> Add MOS solution onLED3, 2 to avoid the light blinked.



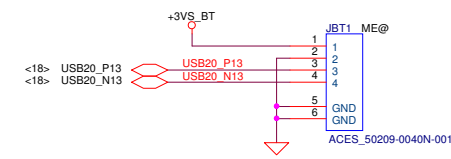
PWR LED HDD LED CapsLK LED



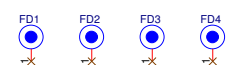
BlueTooth DC



BT Conn.



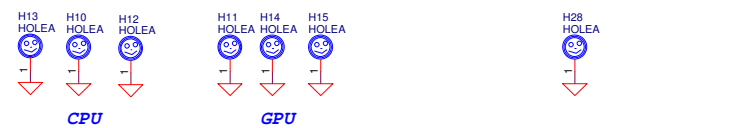
PCB Fedical Mark PAD



Screw Hole

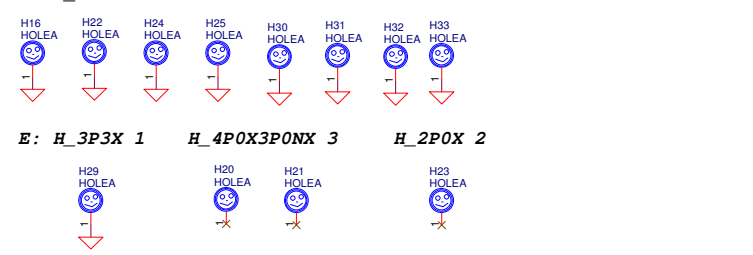
CPU and GPU: H_3P8X 6 MIN PCIE: H_3P3 X 1

C: H_3P8X 3 B: H_3P8X 3 E: H_3P3X 1



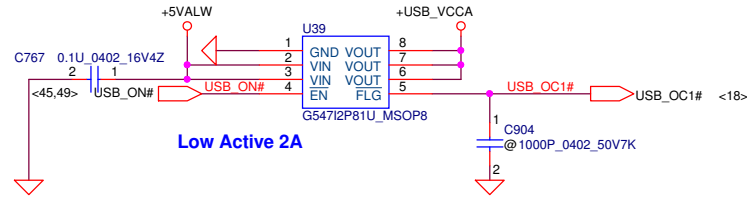
ME: H_8P0 X 8; H_3P3X 1; H_4P0X3P0N X 2; H_2P0X 1

A: H_2P8X 8

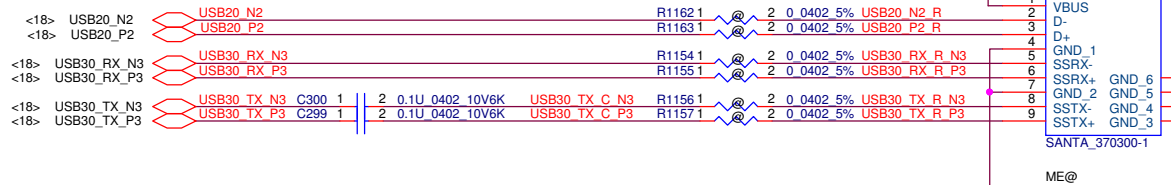
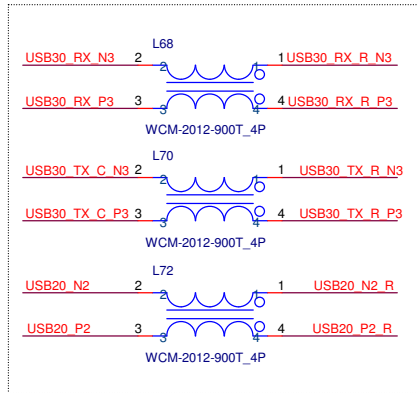


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Issued Date	2011/11/01	Deciphered Date	2012/12/31	Document Number		
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Date:	Monday, January 14, 2013	Sheet	47	of	65	

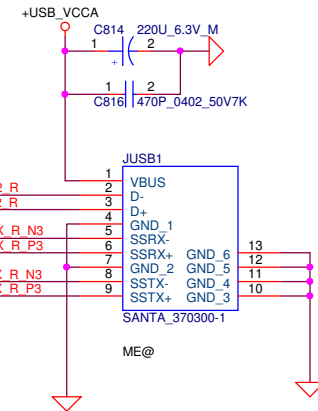
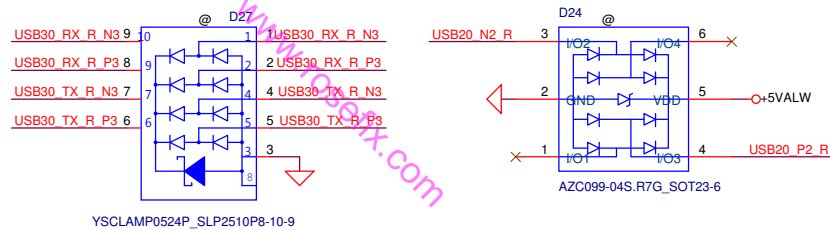
LEFT SIDE USB3.0 PORT X1



For EMI request
USB2.0 choke --> SM070000I00
USB3.0 Choke --> SM070001U00

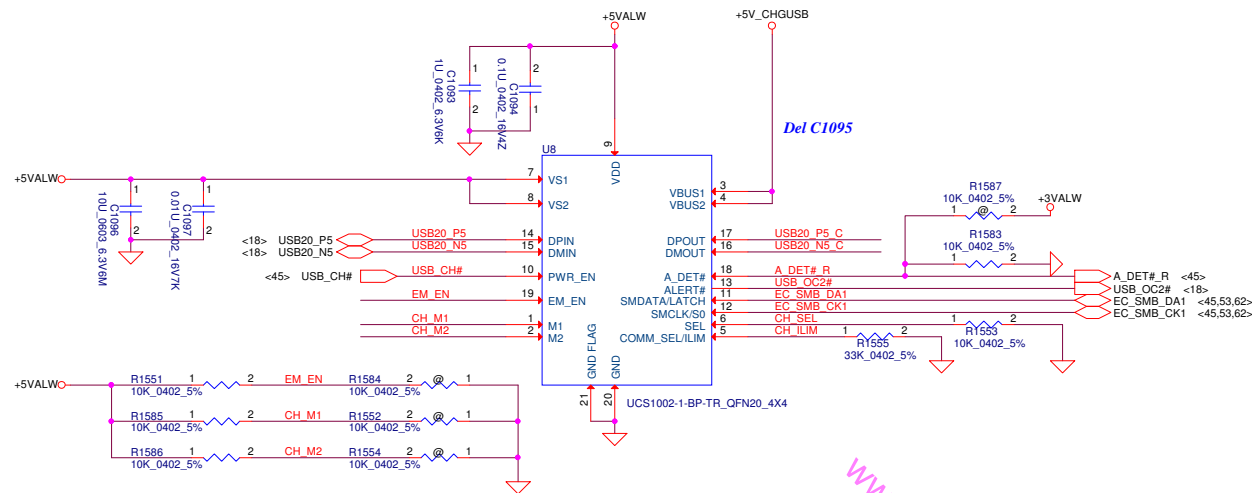


For ESD request



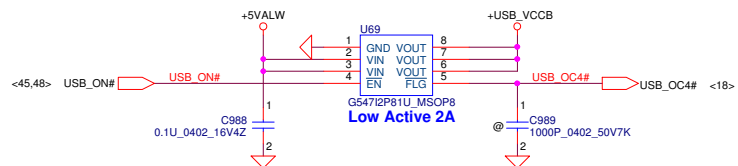
Security Classification		LC Future Center Secret Data				Title					
Issued Date		2011/11/01		Deciphered Date		2012/12/31				USB3.0 PORT	
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Sleep & Charge
Right side USB Charger Port (USB_Port5, near JM1C1)

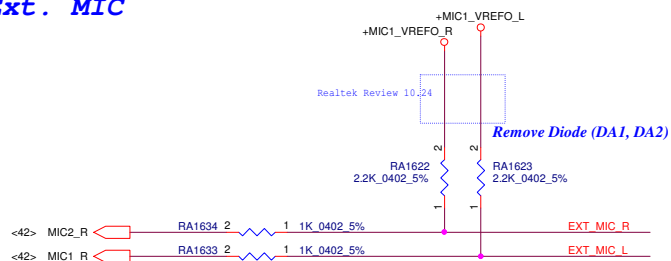


```
2012-0429 --> Set default mode is "BC1.2 CDP" Mode (2.5A on S0) for USB Port5
```

USB Power (USB20_P9)



Ext. MIC



Active Mode Selection:

	M1	M2	EM_EN	ACTIVE	MODE	
0	0	1		Dedicated	Charger	Emulation Cycle
0	0	1		Date Pass	through	
1	0	0		BC1.2 CDP		
0	1	0		Dedicated	Charger	Emulation Cycle
1	1	0		Date Pass	through	
*	1	1		BC1.2 CDP		

ILIM SETTING

Pull Low
OR-500mA
10K-900mA
12K-1000mA
15K-1200mA
18K-1500mA
22K-1800mA
27K-2000mA
★ 33K-2500mA

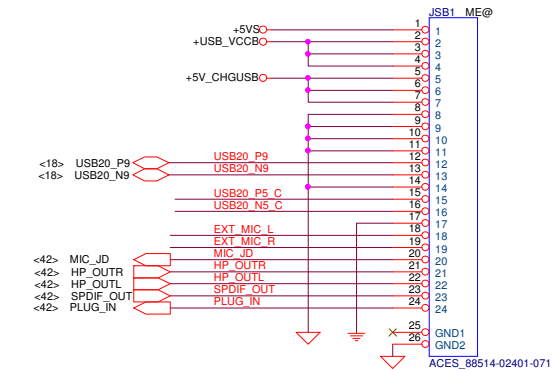
SEL Pin Decode

```

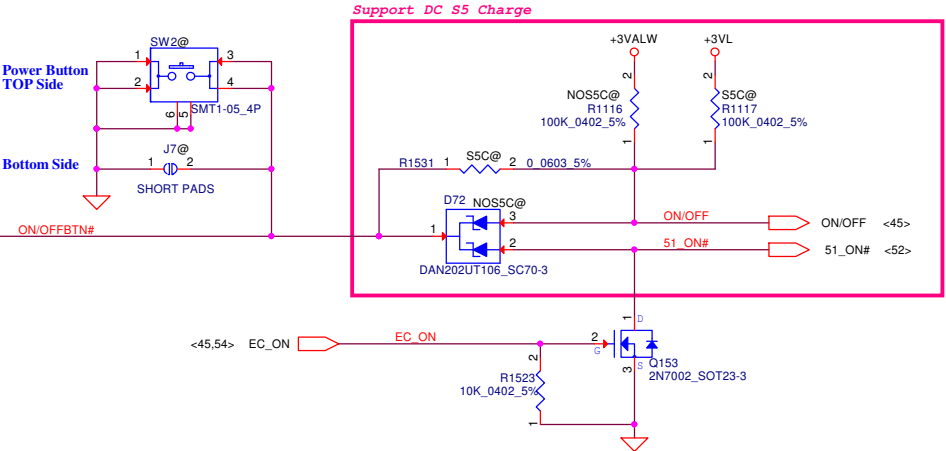
Pull Low
0R -1010_000
★ 10K-1010_000
12K-1010_000
15K-1010_000
18K-0110_000
22K-0110_000
27K-0110_000
33K-0110_000

```

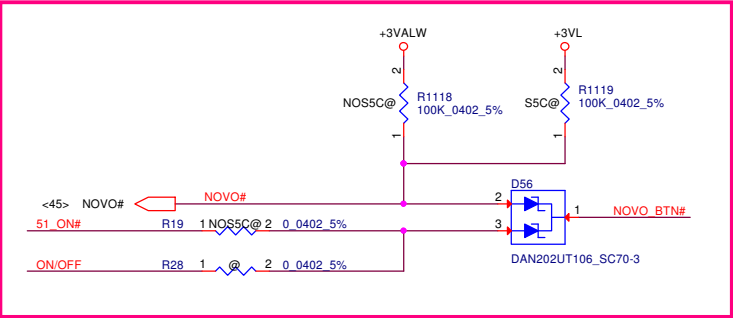
AUDIO/B Conn.



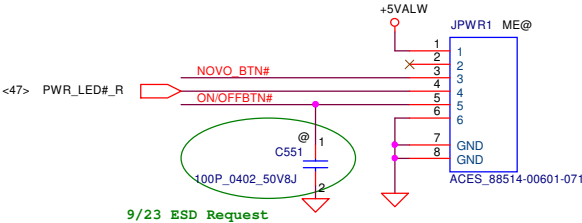
ON/OFF switch



Support DC S5 Charge



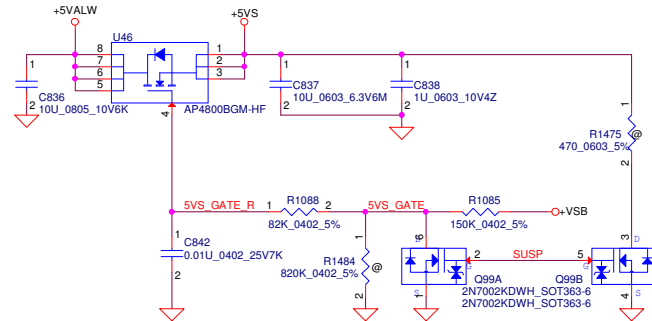
Power Button/B link to Function/B Conn. 10pin



EMI REQUEST 1ST = SCA00000E00
2ST = SCA00000R00

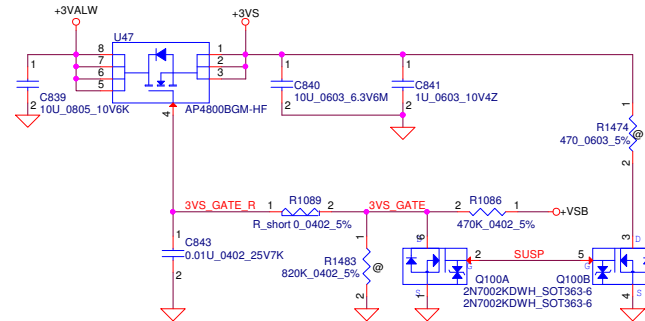
+5VALW to +5VS

AP4800BGM
VGS=1.0V, ID=9A, Rds=18m ohm
VGS=-25V

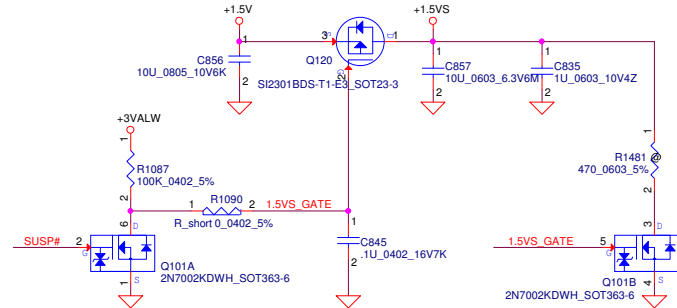
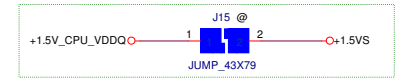


+3VALW to +3VS

AP4800BGM
VGS=1.0V, ID=9A, Rds=18m ohm
VGS=-25V

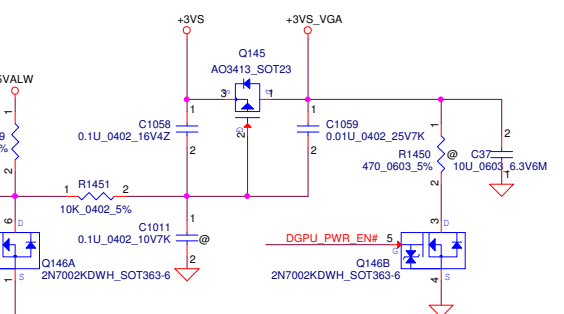
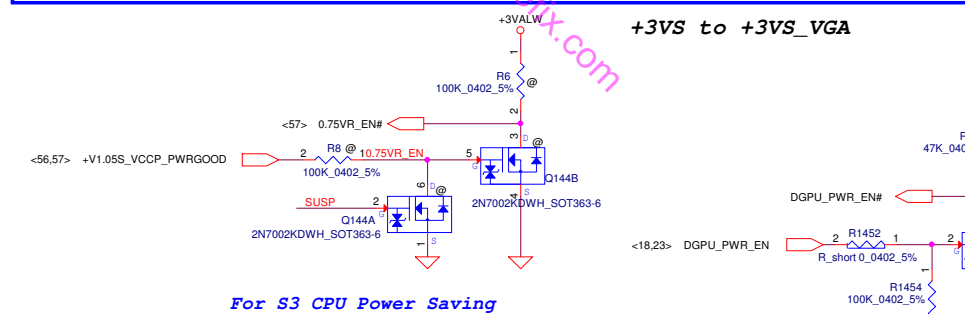
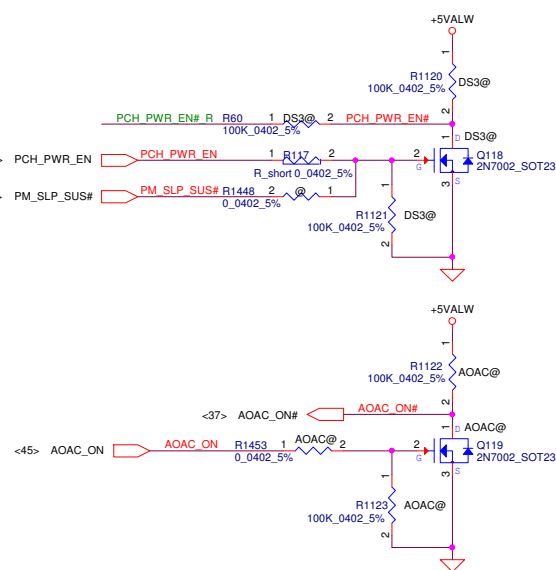
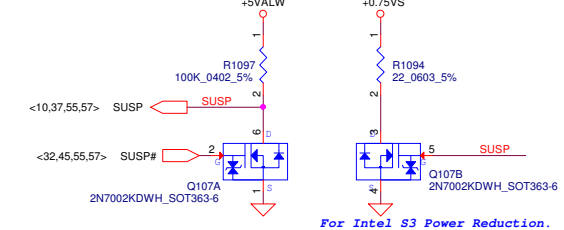
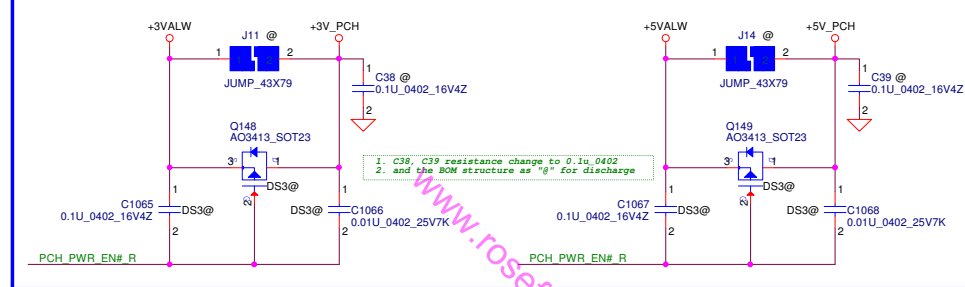


+1.5V to +1.5VS



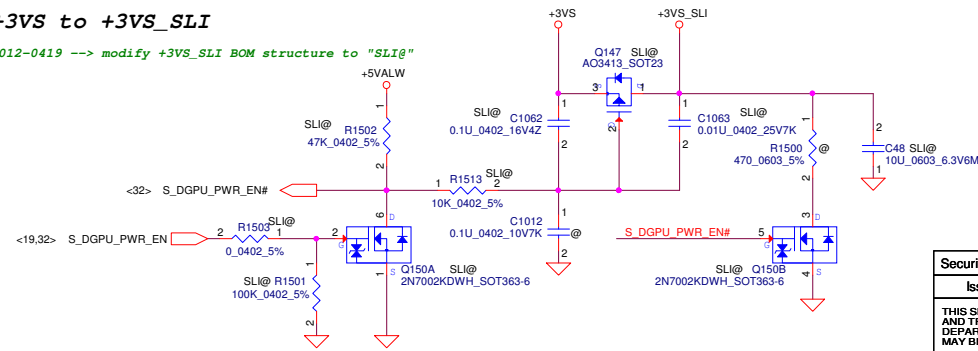
+3VALW to +3V_PCH

+5VALW to +5V_PCH



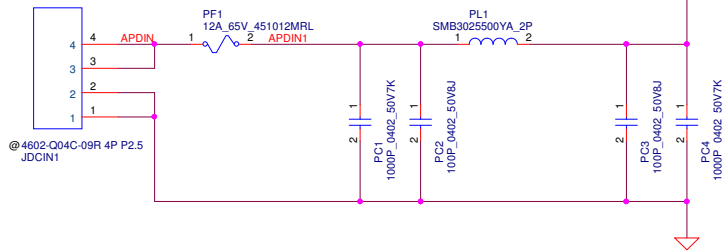
+3VS to +3VS_SLI

2012-0419 --> modify +3VS_SLI BOM structure to "SLI@"

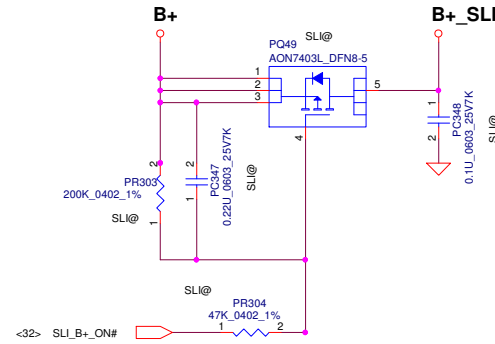


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Date:	Monday, March 11, 2013	Sheet	51	of	65

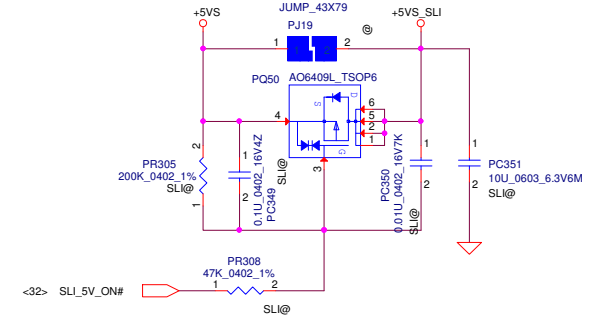
DC030006J00



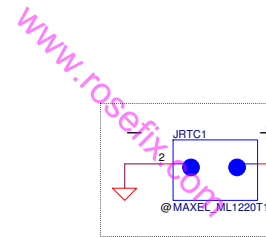
B+ to SLI_B+



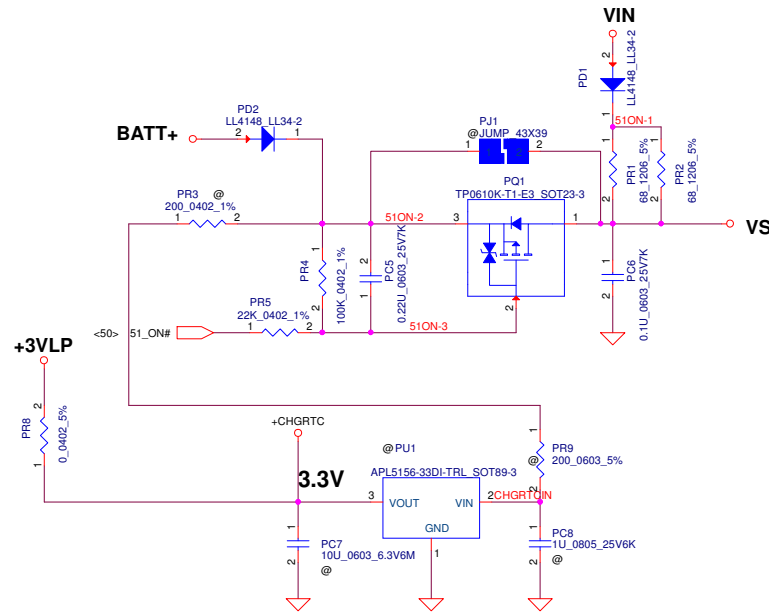
+5VS to +5VS_SLI



2012/04/13
add SLI Hot-plug Load-SW solution

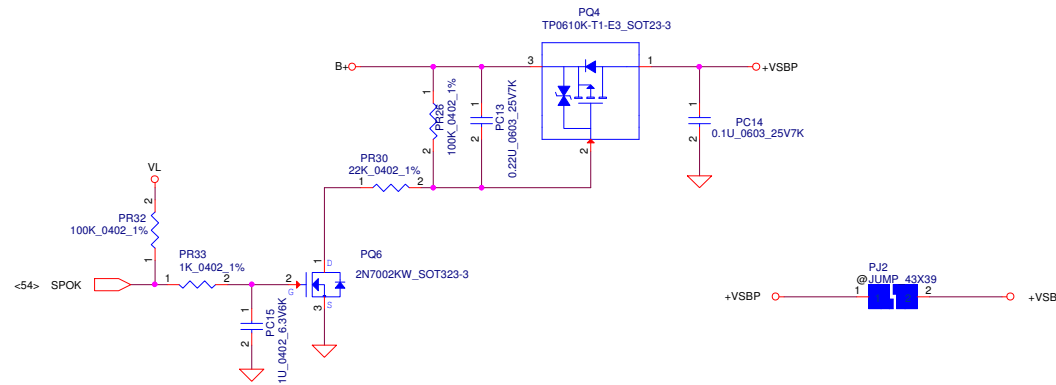
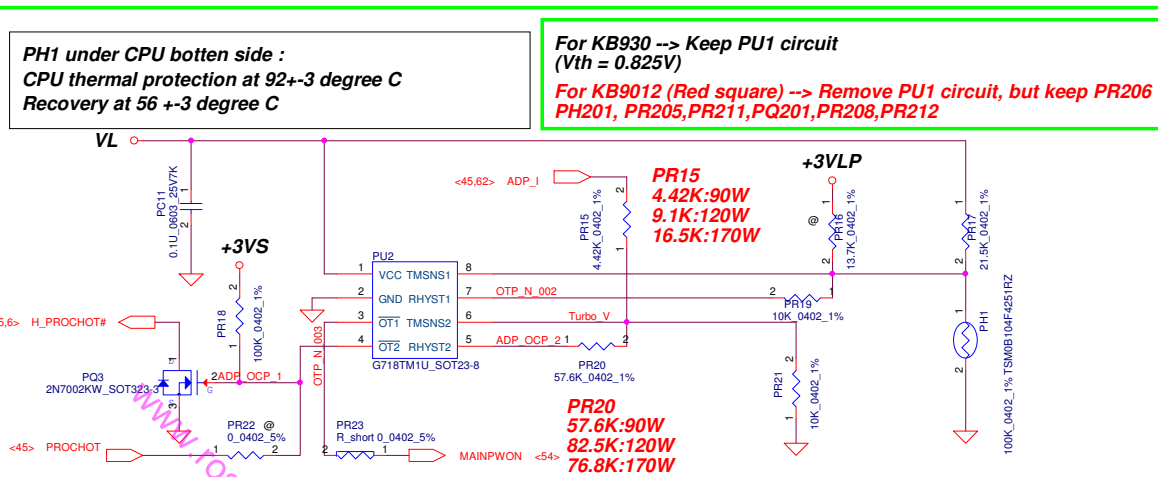
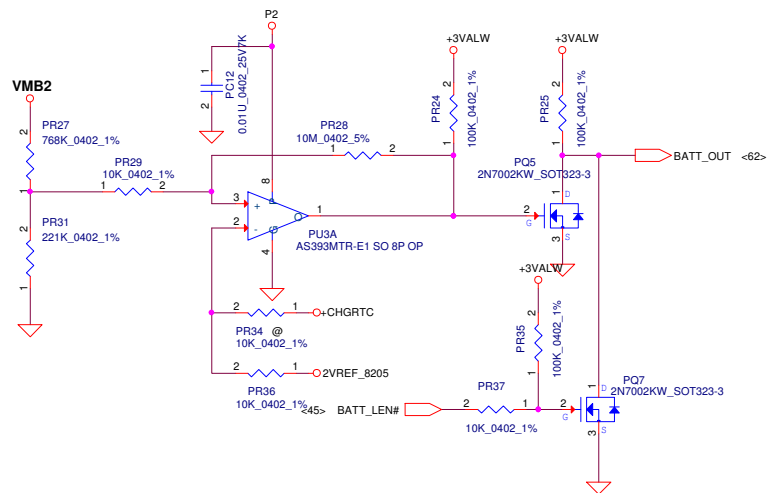
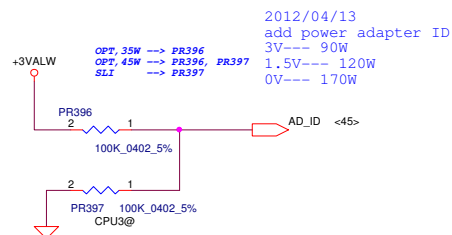
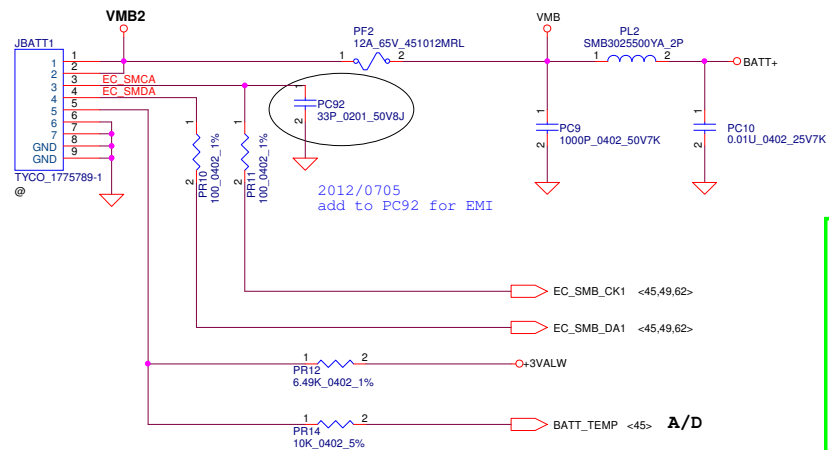


RTC Battery



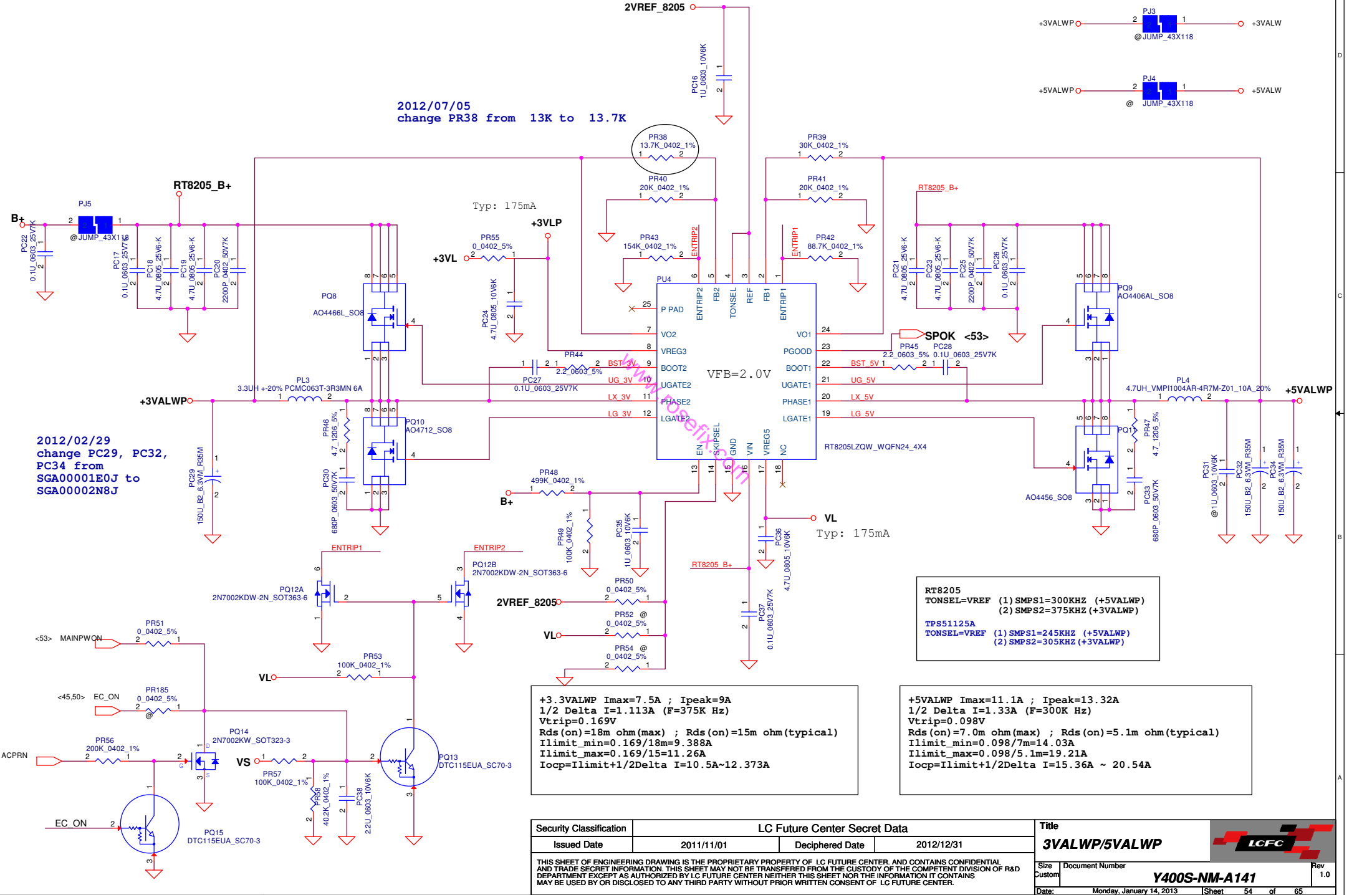
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Issued Date	2011/11/01	Deciphered Date 2012/12/31
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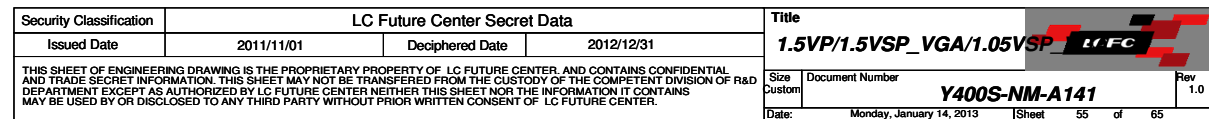
Title	Document Number	Rev
VIN DETECTOR	Y400S-NM-A141	1.0
Date:	Monday, January 14, 2013	Sheet 52 of 65



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Custom					
Date:	Monday, January 14, 2013	Sheet	53	of	65

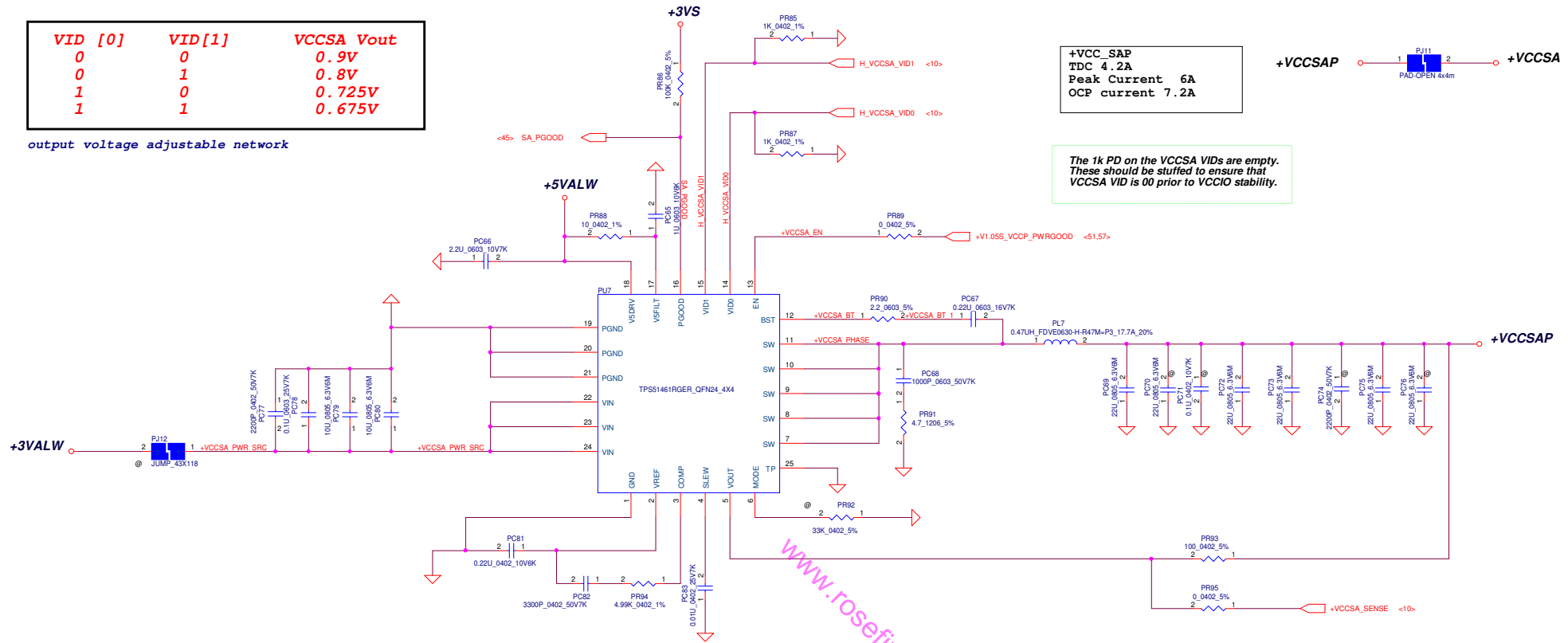
Note:
Use TPS51125 IC can remove RTC refernece LDO
Use TPS51427 IC must keep RTC refernece LDO

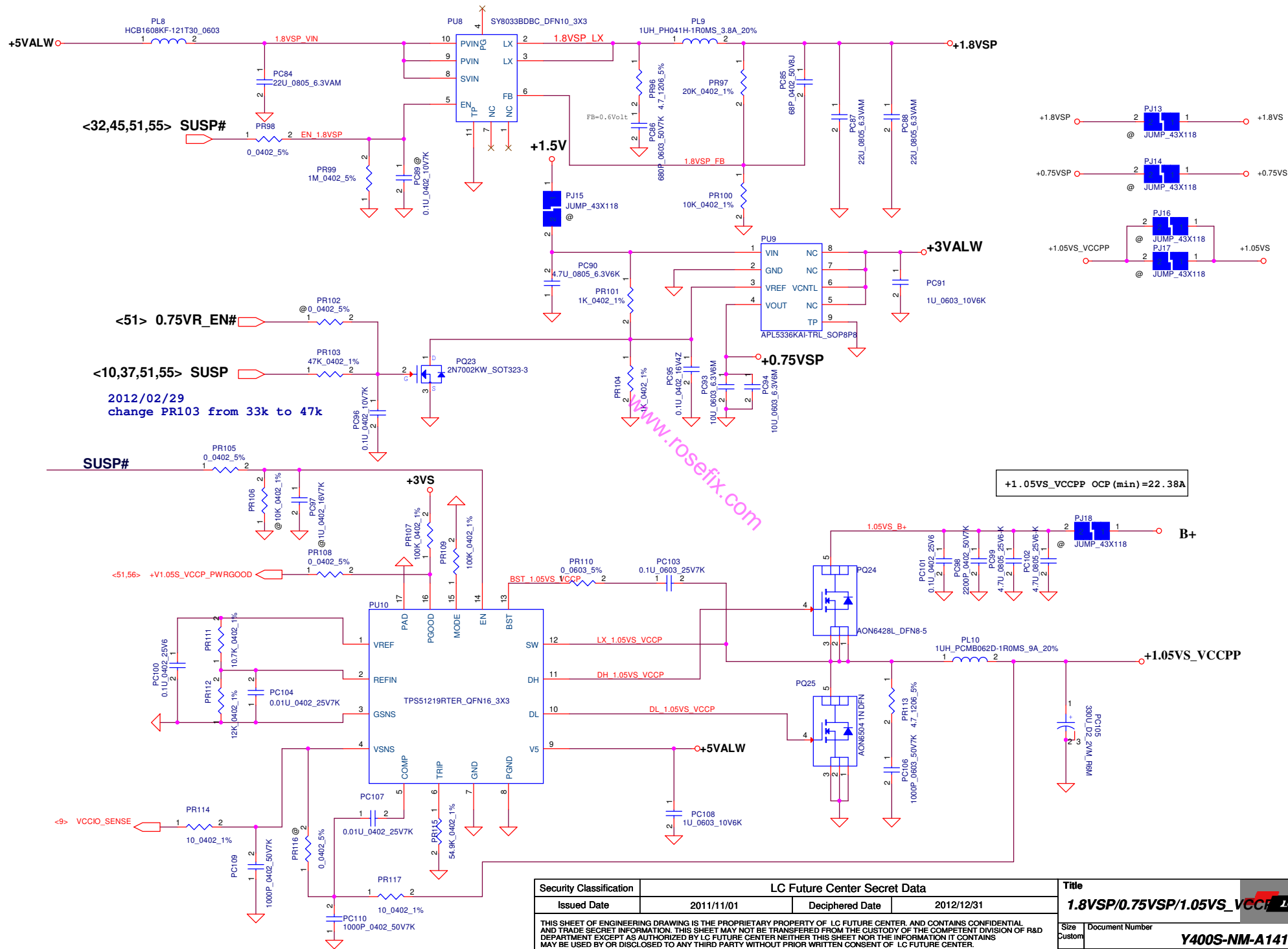




VID [0]	VID[1]	VCCSA Vout
0	0	0.9V
0	1	0.8V
1	0	0.725V
1	1	0.675V

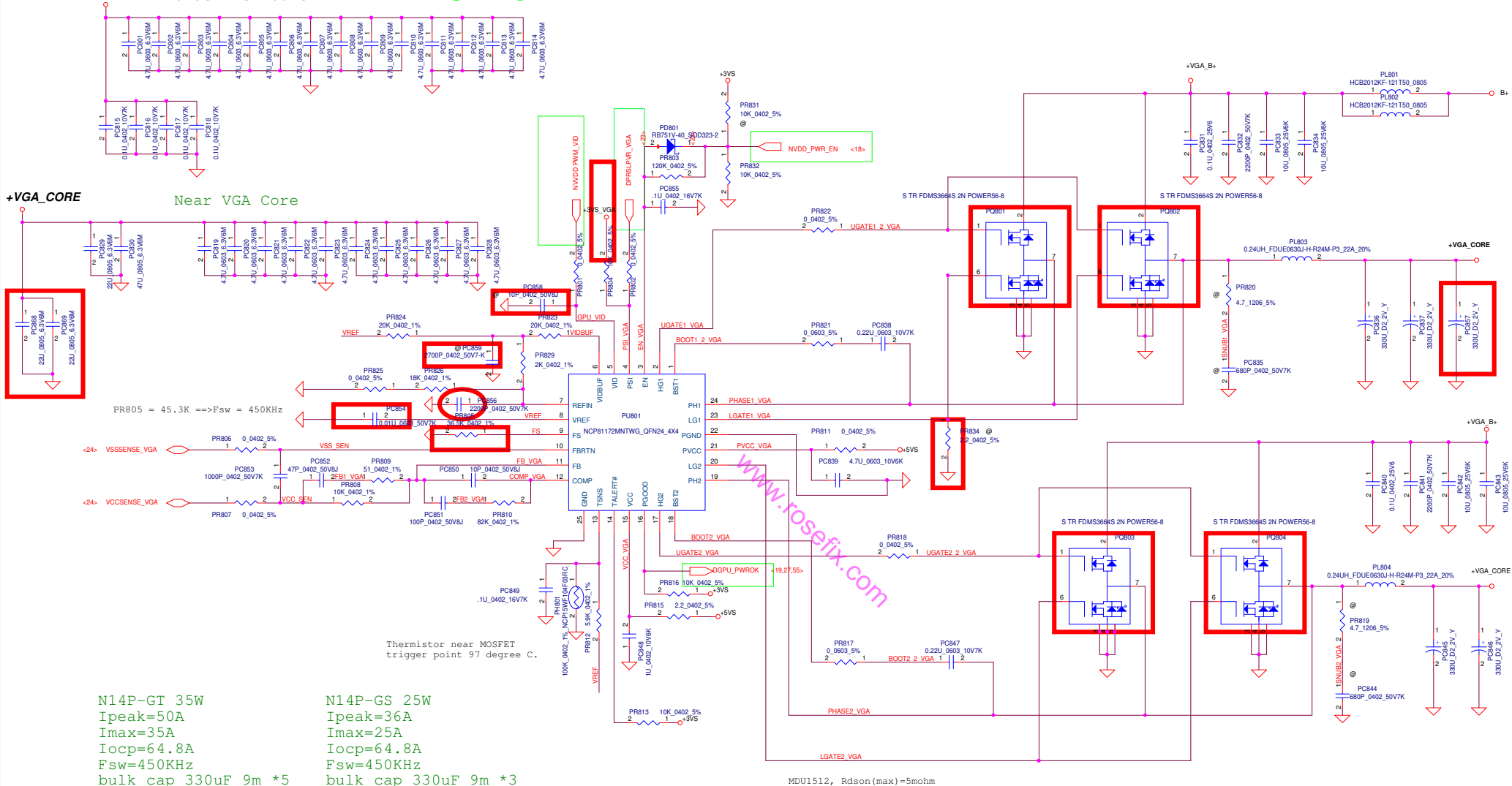
output voltage adjustable network





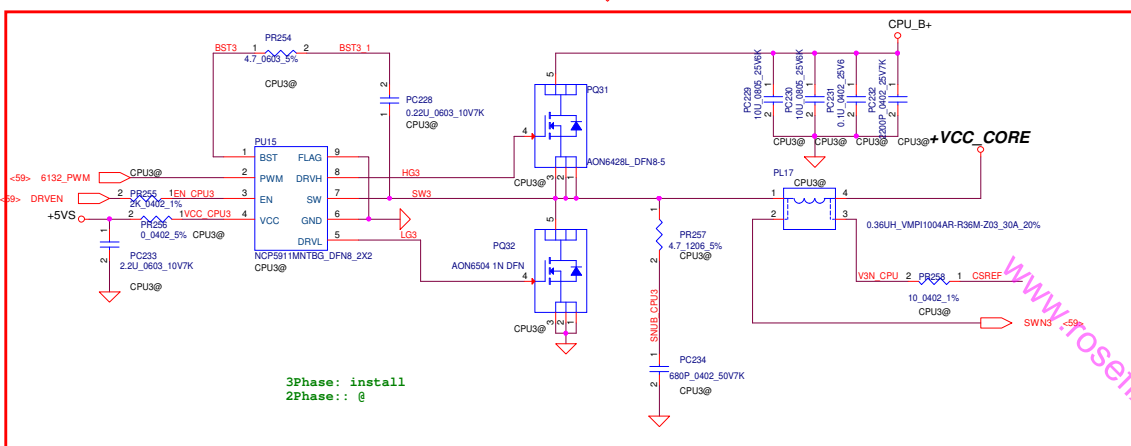
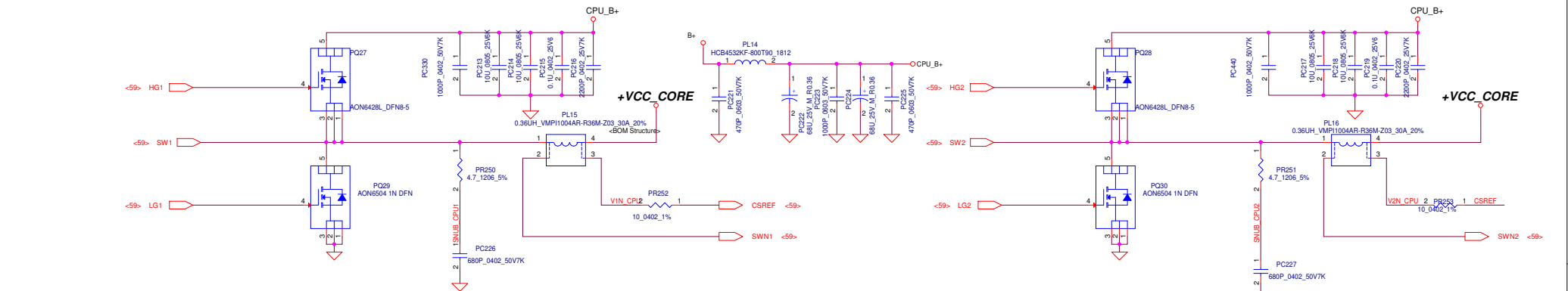
+VGA_CORE Under VGA Core GB4-128 package

+VGA_CORE Near VGA Core



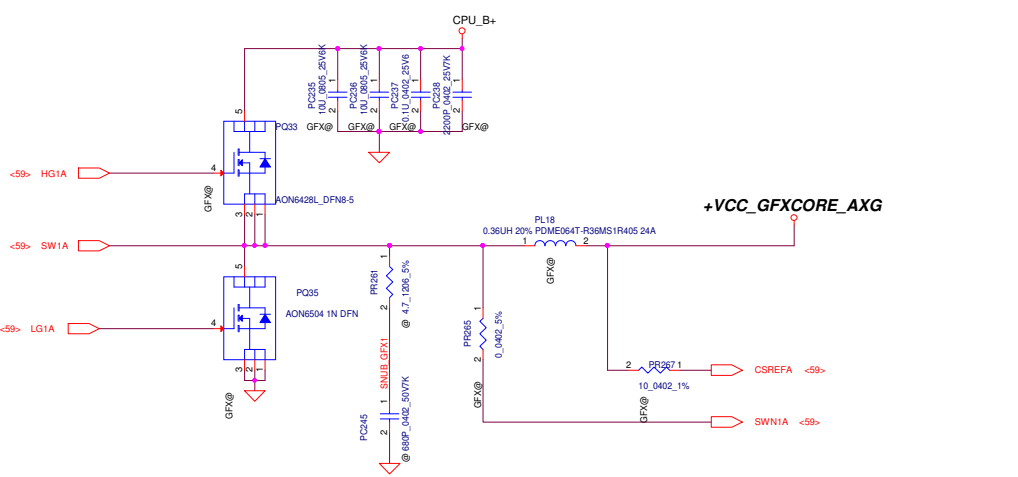
N14P-GT 35W
Ipeak=50A
Imax=35A
Iocp=64.8A
Fsw=450KHz
bulk cap 330uF 9m *5

N14P-GS 25W
Ipeak=36A
Imax=25A
Iocp=64.8A
Fsw=450KHz
bulk cap 330uF 9m *3



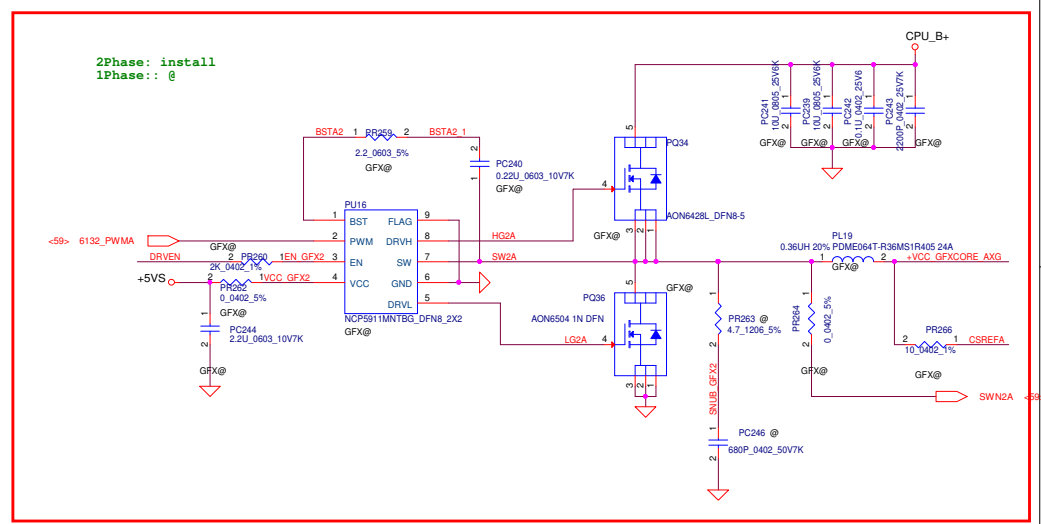
QC 45W CPU
VID1=0.9V
IccMax=94A
Icc_Dyn=66A
Icc_TDC=52A
R_LL=1.9m ohm
OCP-110A

DC 35W CPU
VID1=1.05V
IccMax=53A
Icc_Dyn=43A
Icc_TDC=36A
R_LL=1.9m ohm
OCP-65A



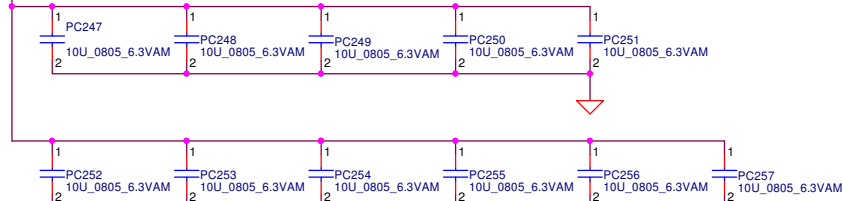
QC 45W GT2
VID1=1.23V
IccMax=46A
Icc_Dyn=33A
Icc_TDC=38A
R_LL=3.9m ohm
OCP-55A

DC 35W GT2
VID1=1.23V
IccMax=33A
Icc_Dyn=20.2A
Icc_TDC=21.5A
R_LL=3.9m ohm
OCP-40A

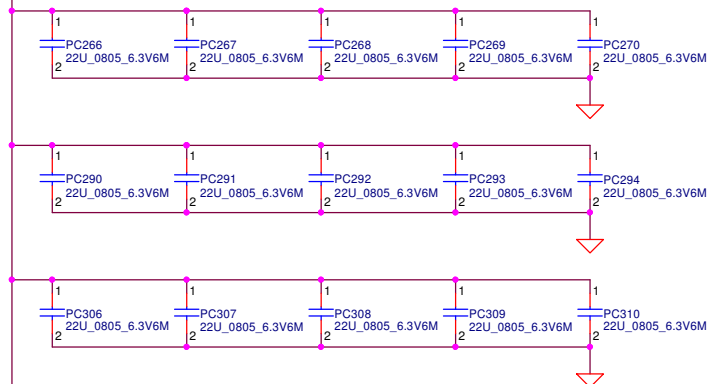


Security Classification	LC Future Center Secret Data		Title	CPU_CORE	
Issued Date	2011/11/01	Deciphered Date	2012/12/31	Size	Document Number
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				Y400S-NM-A141	Rev 1.0
				Date	Monday, January 14, 2013
				Sheet	60 of 65

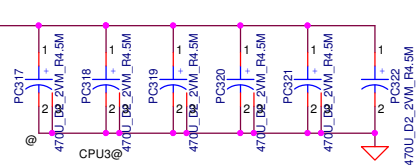
+VCC_CORE



+VCC_CORE



+VCC_CORE



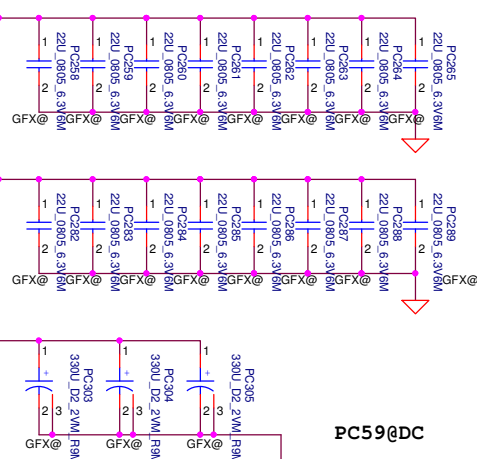
DC: PC73, PC74, PC75, PC76, PC77, PC78 (330uF/9m)
QC: PC76, PC78 (470uF/4.5m), PC73, PC74, PC75 (330uF/9m)

PC8, PC21, PC22, PC63

+CPU_CORE

+VCC_GFXCORE_AXG

+VCC_GFXCORE_AXG



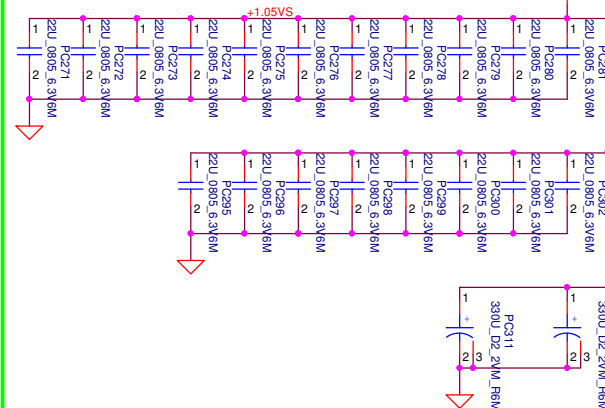
PC38, PC39, PC40, PC41

PC38, PC39, PC40, PC41

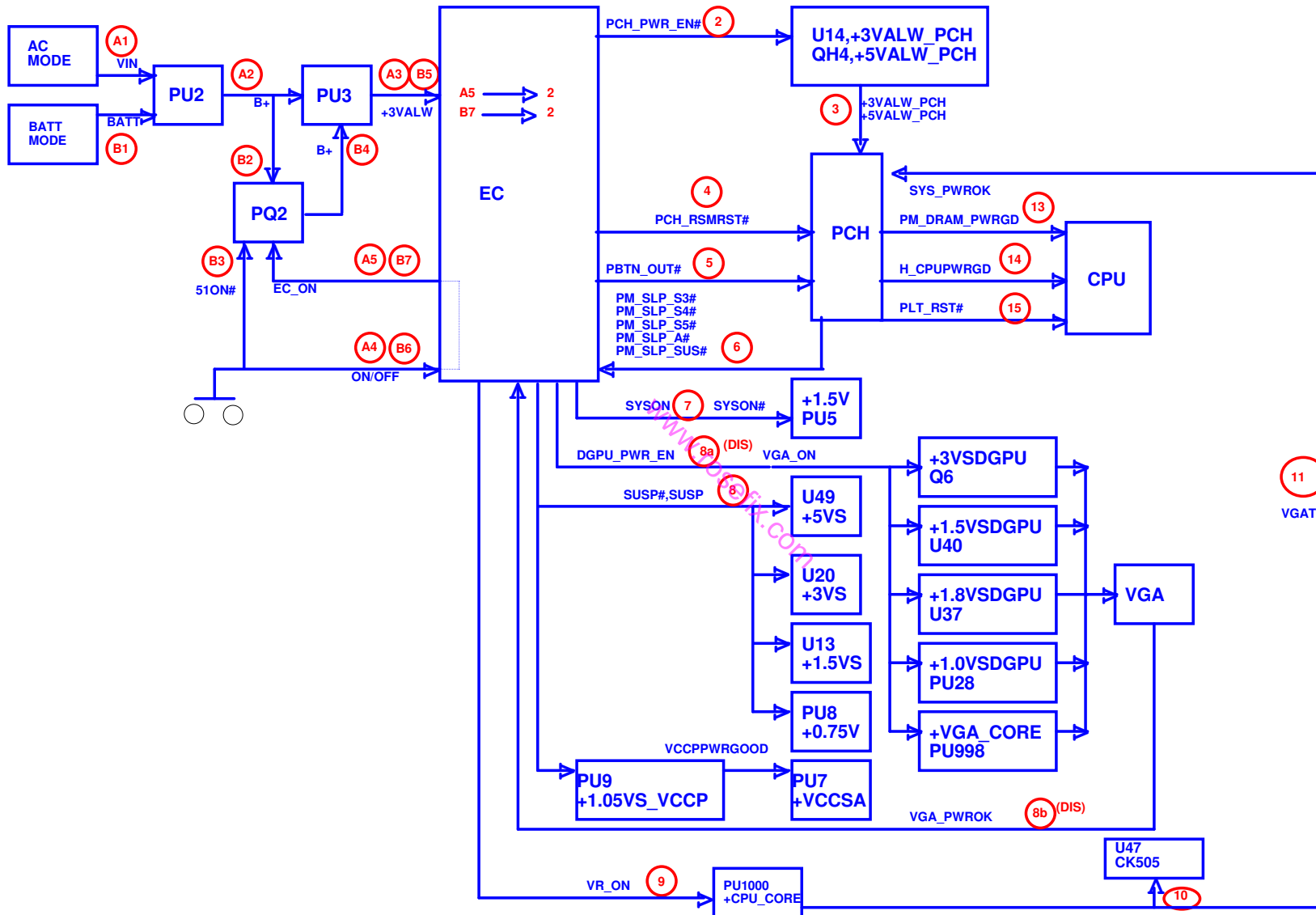
Below is 458544_CRV_PDDG_0.5 Table 5-8.

Socket Bottom	5 x 22 μ F (0805) 5 x (0805) no-stuff sites
Socket Top	7 x 22 μ F (0805) 2 x (0805) no-stuff sites

+1.05VS



PC32, PC49, PC54, PC55, PC56



Version change list (P.I.R. List)

Page 1 of 1
for PWR

Item	Reason for change	PG#	Modify List	Date	Phase
1	For NV suggest	58	Add (reserve parts) PC859		
2	For TI suggest	58	Add (reserve parts) PR834		
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					

Security Classification		LC Future Center Secret Data		Title	
Issued Date		2012/10/11		PIR (PWR)	
Deciphered Date		2014/07/01			
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Size	Document Number	Rev			
Custom		1.0			
Date: Monday, January 14, 2013 Sheet 64 of 65					

QIWY5 HW PIR List

NO	DATE	PAGE	MODIFICATION LIST	PURPOSE
1		P23	Change DGPU_PWR_EN to PLT_RST_VGA#	EVT TO DVT
2		P23	Add CV148	For GC6 function
3				For GC6 function
4				
5				
6				
7				
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14				
14				
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16				

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